

Zoning and the Dynamics of Urban Redevelopment

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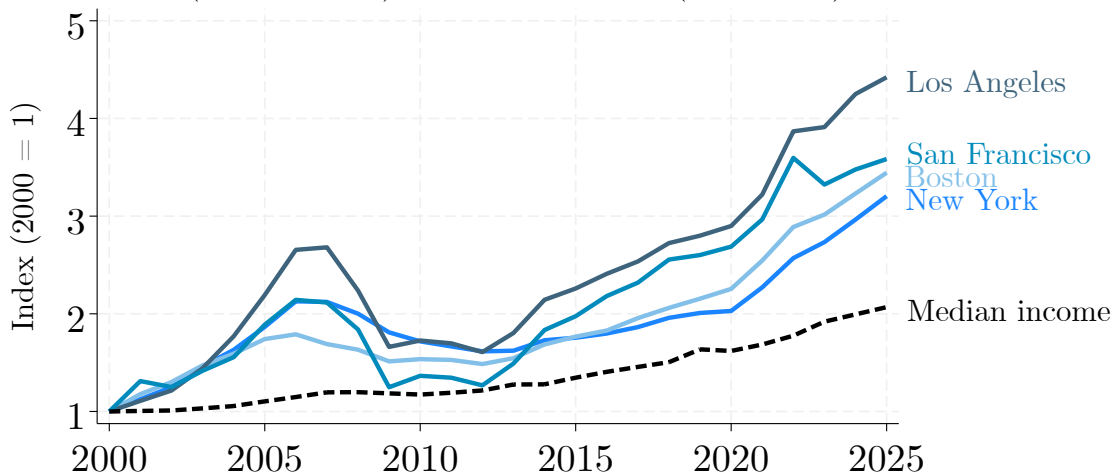
MIT

University of Chicago

January 7, 2026

Home prices have skyrocketed in many large cities

Home prices (selected cities) and median income (nationwide)



Strict zoning is often denounced as a culprit

Zoning: Regulations restricting characteristics of new buildings.

- **Density limits** (e.g., height limits).
- **Use limits** (e.g., residential only).
- Some deregulation (**upzoning**) happening in many cities (Chicago, Cambridge, Minneapolis, Berkeley, Denver, Seattle, Austin, NYC, Portland, Raleigh, San Diego...)

THE WALL STREET JOURNAL.

Only Zoning Reform Can Solve
America's Housing Crisis

Los Angeles Times

L.A. can't become an affordable city
by protecting single-family zoning

FT

FINANCIAL
TIMES

Planning rules are driving the
global housing crisis

The Boston Globe

Do homes really need 2-acre lots?
Spoiler alert: No.

Technological constraints matter too

- In urban cores, vacant land is scarce.
- Change comes from **redevelopment**.
 - Demolishing and replacing old buildings.
- Increasingly how cities grow (Frolking et al., 2024).
- Requires paying large **fixed costs**.



How do zoning and the existing building stock influence urban development?

- First paper to build a **parcel-level land use and zoning panel** for a city.
 - Track all parcels and development events in NYC over 2004–2022.
- **Dynamic spatial equilibrium model** of urban change.
 - **Supply**: Behavior of forward-looking developers given prices and regulation.
 - **Demand**: Spatial equilibrium model predicts how development affects prices.

Policy application: What would be the effects of relaxing zoning?

- **Quasi-experimental evidence** from recent zoning changes.
- Model accurately predicts effects of past upzonings.
- Use model to predict the **effects of relaxing zoning** on **construction** and **prices**.

Findings

- Zoning strongly hinders construction, and relaxing zoning yields **large welfare gains**.
- But **adjustment** to policy change is **slow**, **take-up of upzoning is limited**.
 - Only 1/4 of long-term gains in first decade.
 - Allowing 5 new units yields only 1 built over 40 years.
 - Mechanism: Large **fixed costs** of redevelopment that rise steeply with height.
- **Consequences**: Relaxing zoning most effective where **prices are high/density is low**.
- Affordability **gains geographically spread out** due to migration.

Contribution

City structure and evolution Burgess (1925), Hoyt (1939), Alonso (1964), Mills (1967), Muth (1969), Ogawa & Fujita (1980), Lucas & Rossi-Hansberg (2002), Ahlfeldt, Redding, Sturm & Wolf (2015), Allen, Arkolakis, & Li (2015), Heblich, Redding, & Sturm (2020), Gechter and Tsivanidis (2023), Hsiao (2024)

- Instead of long-run static equilibria, focus on **redevelopment** and **transition path**.
- New granular **data** allows to show new **facts** and build a new **model** of urban change.

Land use regulation Glaeser, Gyourko and Saks (2005), Ganong and Shoag (2017), Glaeser and Gyourko (2018), Hsieh and Moretti (2019), Ospital (2022), Anagol, Ferreira & Rexer (2024), Peng (2023)

- Evaluation of the **effects of relaxing zoning** (reduced-form + credible extrapolation).

Roadmap

- ① Data and context
- ② Empirical evidence on redevelopment and zoning
- ③ Dynamic model of urban change
- ④ Model estimation and validation
- ⑤ Implications for zoning policy

Data and context

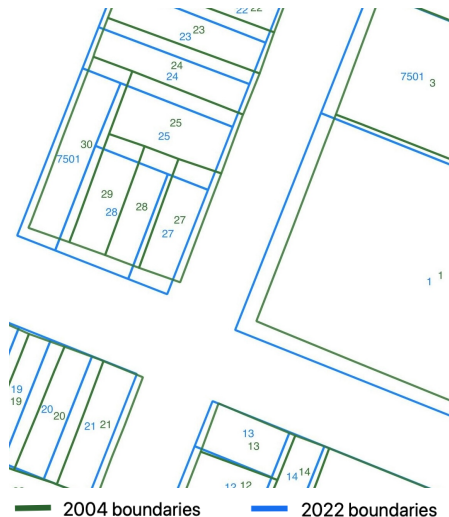
NYC as a case study

- Largest US city (8.5 million inhabitants).
- Wide variation in density and prices.
- Housing affordability crisis.
- Active policy debate.
- Need to gather data on:
 - 1 Floorspace quantities.
 - 2 Floorspace prices.
 - 3 Zoning.



Floorspace quantities: Parcel-level land use

- I use **cadastral maps** to:
 - Partition NYC in time-consistent parcels.
 - Link land parcels to buildings over time.
- Developed a polygon conflation algorithm.
 - 5,700 manual adjustments for a perfect match.
- Track universe of parcels over 2004–2022.
 - 833,000 parcels in total.
- **Building characteristics:**
 - Property tax records (scraping, FOIA requests).
 - StreetEasy (scraped data on 1.6 million units).



Floorspace quantities: Redevelopment events

- Isolate 22,000 **redevelopment events**.
 - Buildings constructed since 2004.
- **Building permits**
- **Certificates of Occupancy**
 - Construction ends.
 - Scraped and digitized 250,000 documents.

The image displays three overlapping examples of New York City Certificates of Occupancy (CO) forms. Each form is issued by the Department of Buildings and certifies that a building complies with applicable laws, rules, and regulations. The forms are for different buildings and locations in New York City.

Top Form: Issued by the Department of Buildings, dated **OCT 22 1992**. The building is located at 1229 HENRY AVENUE, Block 2093, Lot 15, in the 10011 ZIP code.

Middle Form: Issued by the Department of Buildings, dated **JUL 24 2002**. The building is located at 87 ARTHUR ST, Block 348, Lot 6, in the 10011 ZIP code.

Bottom Form: Issued by the Department of Buildings, dated **DECEMBER 15, 2004**. The building is located at 100 WEST 110TH STREET, Block 104, in the 10025 ZIP code.

Bottom Form (NYC Buildings): Issued by the NYC Department of Buildings, dated **11/05/2020**. The building is located at 162 SEVENTH AVENUE SOUTH, Block 00613, Lot 53, in the 10011 ZIP code.

Floorspace quantities: Redevelopment timelines



2014



2015



2017



Building permit issued



Certificate of Occupancy
issued

Floorspace prices

- **Sales prices** from real estate transactions over 2003-2022 (1.2 million observations).
 - From NYC Department of Finance.
- Yearly **rent** measures for 73,000 buildings.
 - Scraped 600,000 tax documents.
- Digitized historical sales data from the Real Estate Board of New York.
 - 63,000 sales over 1950–1985.

DETAILED VALUE INFORMATION		
Property Address: 30 AVENUE B		Borough: MANHATTAN
		Block: 398 Lot: 32
Building Class: C7 - Walk-up apartments		
Market Value: Finance multiplied your gross income by the gross income multiplier to determine the market value of your property. Any difference between your calculation and that of the Department of Finance is due to rounding.		
• The Department of Finance estimates that as of January 5, 2010, the market value for this property is \$1,200,000. Finance will use this market value to determine your property taxes starting July 1, 2010. • Finance estimated your property's market value using the income approach.		
Factors Used By Finance To Determine Market Value:		
• Building Gross Square Footage: We estimated building gross square footage at 8,700. • Gross Income: We estimated gross income at \$258,000.		

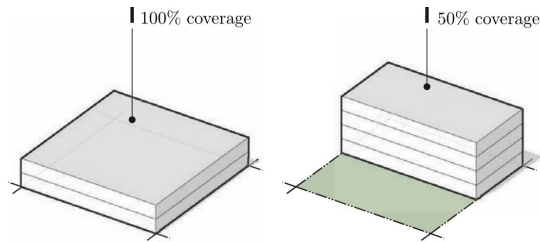
Borough: 1 Block: 788 Lot: 78	
Tax Class: 4 Building Class: L1 Units: 123 non-residential	
ABOUT YOUR PROPERTY TAXES	
Property taxes are determined using a complex formula that takes into account many different amounts and calculations. Visit www.nyc.gov/nopv for more information about property valuation and taxation.	
The Department of Finance estimates that as of January 5, 2019, the Market Value for this property is \$40,925,000. The Department of Finance will use this Market Value to determine your property taxes starting July 1, 2019.	
The Department of Finance estimates your property's Market Value using the income approach. Market Value is determined by dividing the net operating income by the overall cap rate.	
The following factors are used by the Department of Finance to determine Market Value:	
Estimated Building Gross Square Footage: 154,516	
Estimated Gross Income: \$7,211,072	

Zoning

- I match my land use panel with zoning regulations for each year.
- Zoning limits the **size** and **use** (residential vs. commercial) of new buildings.
- Main zoning instrument in NYC: limits on the **Floor Area Ratio** (FAR).

$$\text{FAR} = \frac{\text{sq. ft of floorspace}}{\text{sq. ft of land}}$$

Buildings with a FAR of 2

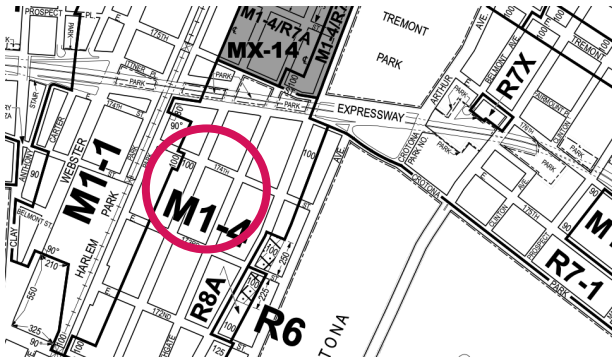




Zone R1-2

- **Residential** allowed up to 0.5 FAR points.
- **Commercial** not allowed.

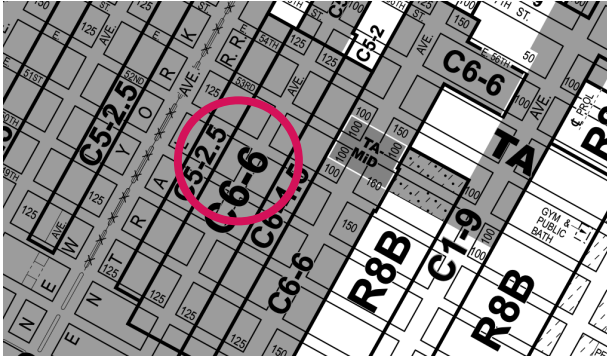




Zone M1-4

- **Residential** not allowed.
- **Commercial** allowed up to 2 FAR points.





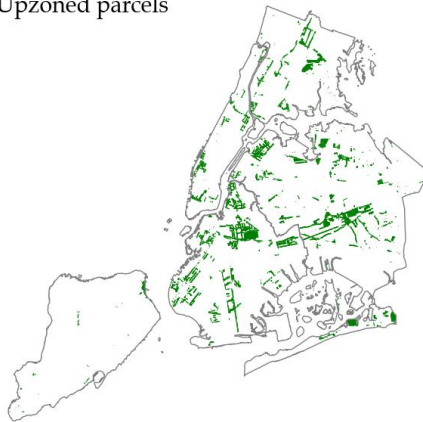
Zone C6-6

- **Residential** allowed up to 10 FAR points.
- **Commercial** allowed up to 15 FAR points.



- Zoning ordinance adopted in 1961.
 - Promoted “towers in the park.”
 - Separated residential/commercial uses.
 - Goal was to limit negative externalities.
- Strong persistence over time.
- Several areas upzoned in past decades.
 - Allows to evaluate the effects of upzoning.
- Focus on large planner-initiated changes.
 - Many veto players.
 - Exact changes and timing unclear when upzoning discussions begin.
 - Leveraged in reduced-form.

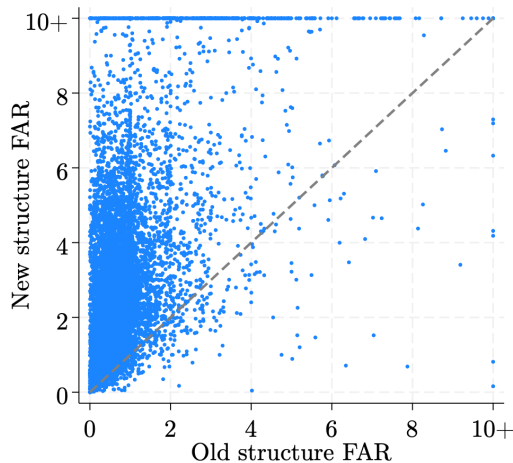
■ Upzoned parcels



Empirical evidence

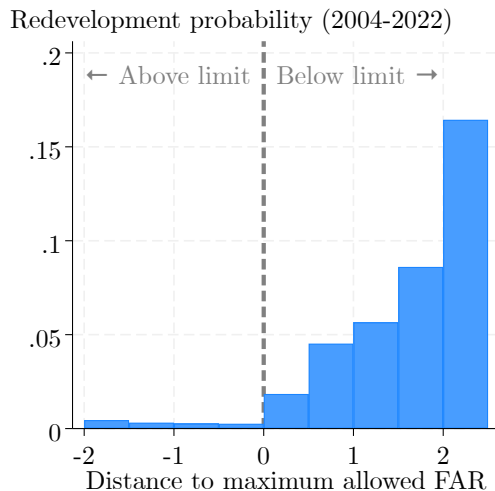
Redevelopment means densification

- New buildings 3.4 times larger than the ones they replace (on average).
- Suggests large redevelopment **fixed costs**.



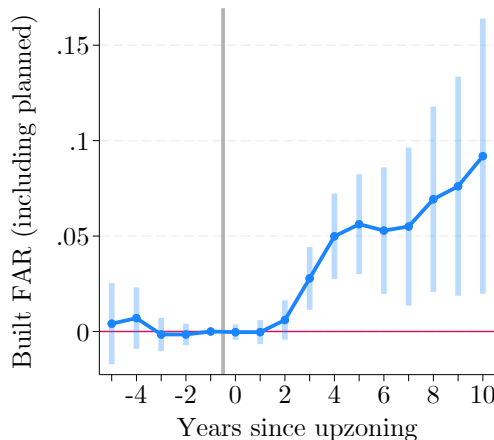
Redevelopment mostly happens when upward growth is allowed

- Buildings at/over the zoning limit are seldom redeveloped.



The effects of upzoning on built FAR materialize slowly

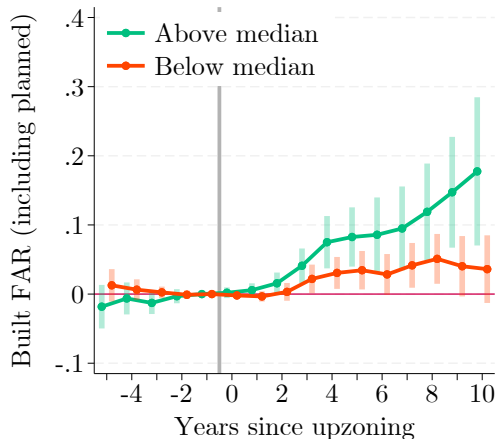
- Compare parcels upzoned earlier vs. later.
 - Identifying assumption: Exogenous timing.
(de Chaisemartin & d'Haultfœuille, 2020)
- Upzoning increases allowed FAR by ≈ 1 .
- $\approx 10\%$ of increase used after a decade.



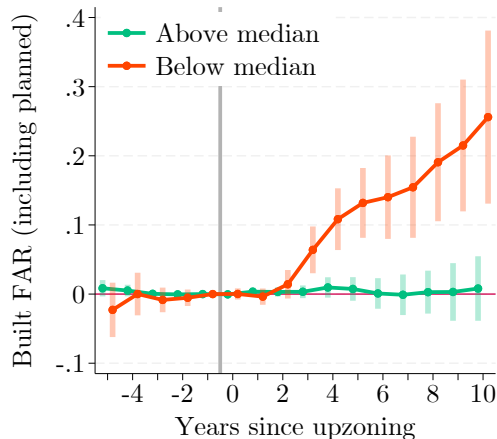
The effects of upzoning vary widely

Effects concentrate in high-price neighborhoods and underbuilt parcels

Heterogeneity by baseline neighborhood floorspace prices

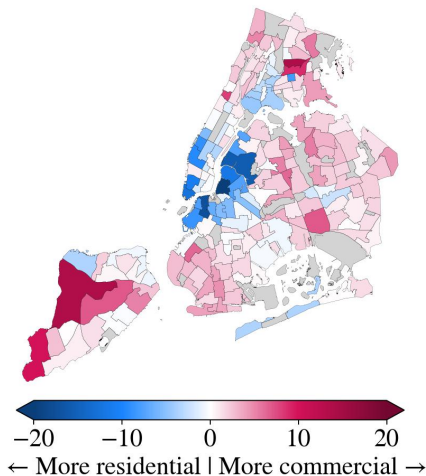
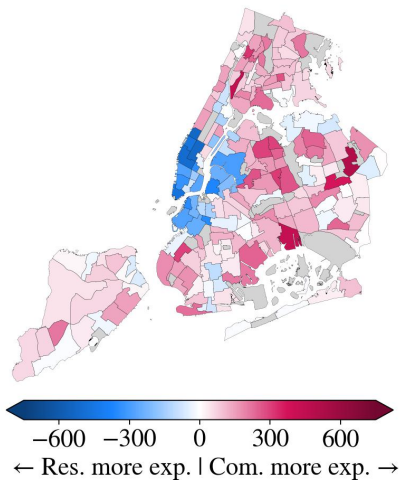


Heterogeneity by baseline built FAR (within neighborhood)



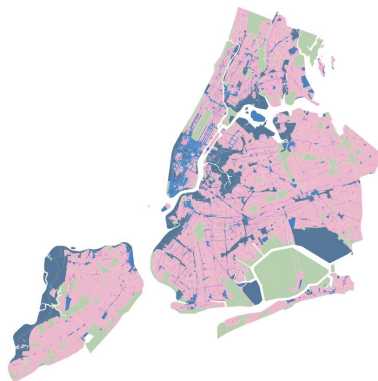
Redevelopment reallocates land to its most profitable use

Commercial-residential price diff. (\$/sq. ft) 2004-2022 change in commercial share (pp)



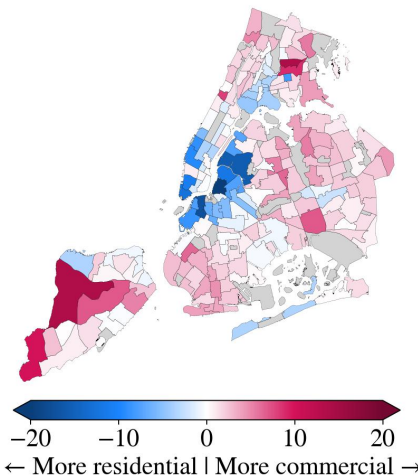
Zoning constrains the reallocation of land uses

Zoning map



Residential Manufacturing
Commercial Parks

2004-2022 change in commercial share (pp)



Remaining questions

① Extrapolation:

- Upzoned neighborhoods were selected: what would happen if we upzoned **elsewhere**?
- The upzonings were limited in scope: what would happen if we upzoned **more**?
- Effects take time to materialize: how large are they in the **long run**?

② What are the effects of allowing more construction on **prices**?

- Answer these questions with a **model of urban change** incorporating:
 - Dynamic transition path.
 - Fixed costs of redevelopment increasing with old building FAR.
 - Use (residential/commercial) and bulk (FAR) zoning constraints.
 - Externalities associated with different land uses.
- Use reduced-form analysis of upzoning for **model validation**.

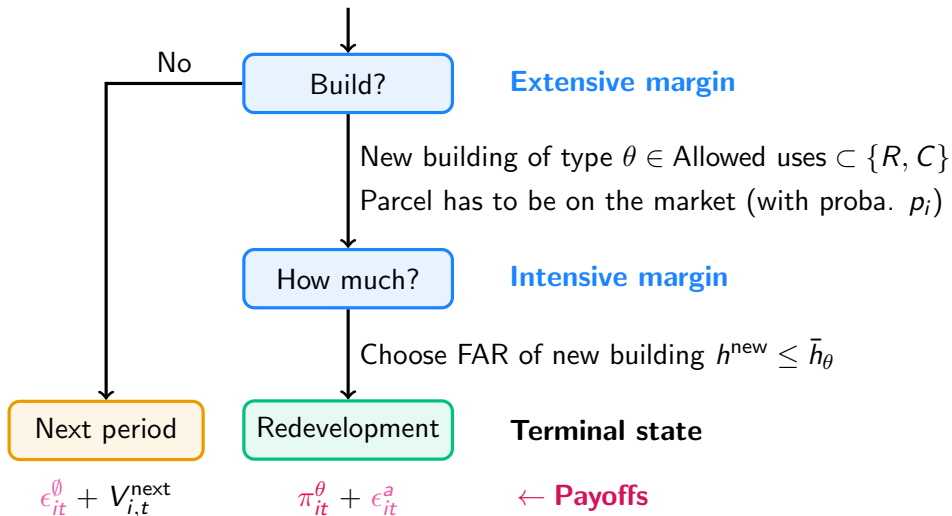
A dynamic model of urban change

Supply of floorspace

Dynamic behavior of developers given prices and regulation

Developer choices

Developers are atomistic and perfectly foresighted. They make decisions each year:



Developer profit function

$$\pi_{it}^{\theta} = \max_{h^{\text{new}} \leq \bar{h}_{\theta it}} \left[\underbrace{P_{\theta it}^{\text{new}}(h^{\text{new}}) - P_{it}^{\text{old}}}_{\text{Change in property value}} - \underbrace{[VC_{it}(h_{it}^{\text{new}}) + FC_{it}]}_{\text{Cost of redevelopment}} \right]$$

- As buildings age, P_{it}^{old} decreases, redevelopment becomes more profitable.
- **Variable costs (VC)**: construction costs, increase with size of new building.
- **Fixed costs (FC)**: eviction, demolition, permitting, etc.
 - Vary by building (e.g., larger for bigger buildings in dense neighborhoods).

Demand for floorspace

Evolution of prices given developers' choices

Demand for floorspace

- Neighborhoods endowed with commercial and residential floorspace.
- Workers rent housing H , choose home and work locations (n, m) each period:

$$\max_{n, m, H, C} \underbrace{B_{nt}}_{\text{amenities}} \cdot \underbrace{(H - \underline{H}_{nt})^\beta}_{\text{housing}} \cdot \underbrace{C^{1-\beta}}_{\text{other cons.}} \cdot \underbrace{d_{nmt}^{-1}}_{\text{commuting}} \cdot \underbrace{z_{nt}^H z_{mt}^W}_{\text{idiosyncratic preferences for home/work locations}}$$

$$\text{s.t. } C + R_{nt}H \leq \text{Income}_t$$

- Labor income = $s \cdot w_{mt}$, skill s lognormally distributed.
- Income from rented floorspace redistributed proportionally to labor income.
- **Key extension:** Subsistence level of housing and realistic distribution of worker skills.
 - Essential to rationalize consumed quantities of housing.

Amenities and productivities

- Firms consume commercial floorspace:

$$Y_{jt} = A_{mt} \cdot \underbrace{H_{Fmt}^{\alpha_m}}_{\text{Commercial floorspace}} \cdot \underbrace{L_{Fmt}^{1-\alpha_m}}_{\text{Labor}}$$

- Amenities (B) and productivities (A) vary with density of residents (L_R) and jobs (L_F).

$$B_{nt} = \bar{B}_{nt} L_{Rnt}^{\gamma_{RR}} L_{Fnt}^{\gamma_{CR}} \quad A_{mt} = \bar{A}_{mt} L_{Rmt}^{\gamma_{RC}} L_{Fmt}^{\gamma_{CC}}$$

→ γ are the agglomeration elasticities (could be positive or negative).

- City population and congestion increase with expected utility.

Dynamic spatial equilibrium

An equilibrium is a path of

- Local floorspace supplies, rents, and prices;
- Population and job distributions;
- Local productivities, amenities, wages;
- Commuting costs;

Such that:

- ① Developers maximize profits.
- ② Workers maximize utility, firms maximize profits.
- ③ Productivities, amenities, and commuting costs adjust (agglomeration/congestion).
- ④ Floorspace markets clear.
- ⑤ Prices are the discounted sum of rents.

Model estimation and validation

Supply-side parameters

Prices

- Value of a building:

$$P_{\theta it} = \underbrace{\rho_{nt}^{\theta}}_{\text{Location premium}} \times \underbrace{Q(\mathbf{x}_{it})}_{\text{Building quality}} \times \underbrace{h_{it}}_{\text{Building size}}$$

- Estimated through a hedonic regression on the sales data.
- Focus on buildings at/above FAR limit (option value of redevelopment ≈ 0).
- Quality proxied using rich building characteristics in $\mathbf{x}_{it}$ (e.g., age, grade, materials, FAR).

Variable costs estimation

- **Identification approach:** Revealed preferences.
 - Use data on FARs chosen by developers for new buildings + prices/zoning they faced.
- **Challenge:** Zoning induces wedge between price and marginal cost of floorspace.
- **Parametric assumption:** Cobb-Douglas production of floorspace with capital and land.

$$VC_{it}(h_{it}) = \alpha_{bt}^{\theta} h_{it}^{1/\zeta} e^{\eta_{it}}$$

- Developers choose FAR $h \leq \bar{h}$ to max profits: $\max_{h \leq \bar{h}} \{P(h) - VC(h)\}$.
 - Use variation on chosen FAR when below the limit and whether the limit is hit.
 - Recover parameters through **maximum likelihood estimation**.
- I find costs rise steeply with FAR and align with engineering estimates/reported costs.

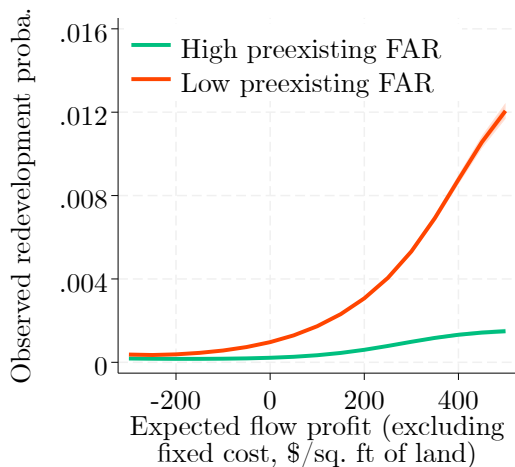
Fixed costs estimation: Intuition

- **Idea:**

- Compute expected profit from redevelopment, excluding fixed costs.
- Find fixed costs to best match relationship with redevelopment probability.

- **Parameterization:** fixed costs

- Increase with size of old building.
- Increase with neighborhood density.
- Larger in historic districts.



Fixed costs estimation: Algorithm

- Requires solving the full model.
 - Option value of redevelopment depends on future path of prices.
 - This path depends on the collective behavior of developers, which depends on fixed costs.
 - Assume that developers believe fundamentals and zoning will stay constant after 2019.
- Estimation using **full-solution approach** (extending Rust, 1987).
 - Inner loop: Estimate developers' value function.
 - Middle loop: Recover price paths by repeatedly solving the demand model.
 - Outer loop: Maximum likelihood.
- I find fixed costs increase sharply with existing buildings' FAR.

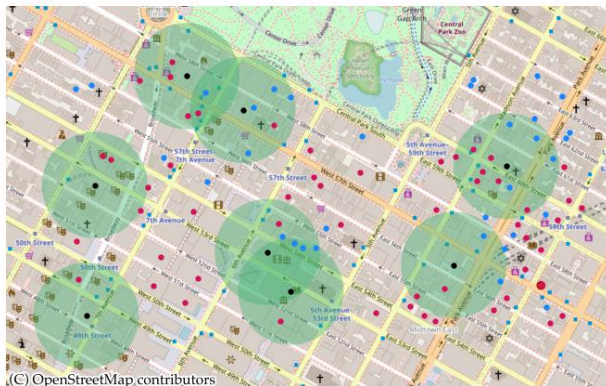
Demand-side parameters

Demand model estimation

- Fundamentals estimated to match commuting flows, rents, and floorspace supplies.
→ Data: LEHD LODES.
- Commuting elasticity estimated using the correlation between commuting flows and commuting times.
→ Data: Google Maps.
- Migration and congestion elasticities calibrated from literature ($\varepsilon_M = 3$, $\varepsilon_M = 0.15$).
- Agglomeration externalities estimated by measuring local demand elasticities.

Price effects of new construction

- Draw 500-ft disks around large construction events (290 residential, 112 commercial).
- Compare the evolution of rents in disks treated earlier vs. later.



Events (new residential buildings)

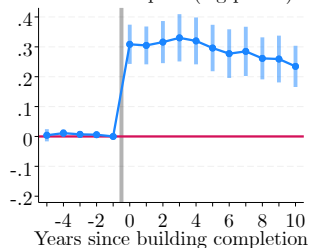
- New buildings
- 500 ft buffers

Buildings with available rent data

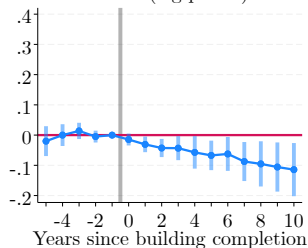
- Residential
- Commercial

Effects of residential construction

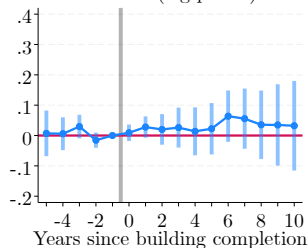
Residential floorspace (log points)



Residential rents (log points)

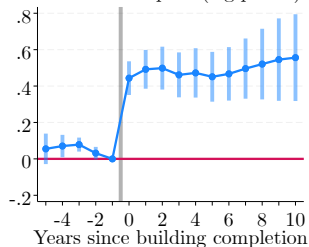


Commercial rents (log points)

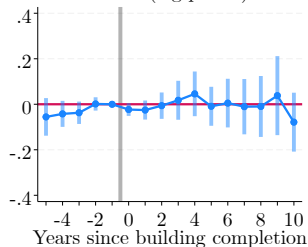


Effects of commercial construction

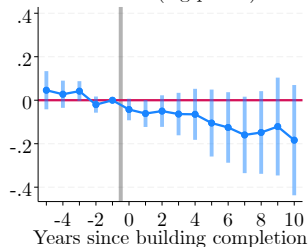
Commercial floorspace (log points)



Residential rents (log points)



Commercial rents (log points)



Price effects of new construction

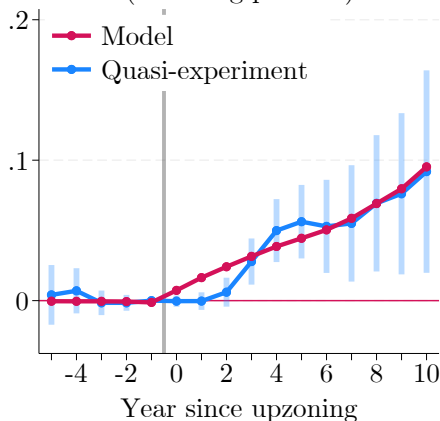
- Estimate agglomeration parameters via **indirect inference** to match reduced-form.
- I find positive agglomeration externalities within uses, limited externalities across uses.
 - Effect of residents on other residents: $\gamma_{RR} = 0.11$.
 - Effect of firms on other firms: $\gamma_{CC} = 0.07$.
 - Corresponding estimates in Ahlfeldt et al. (2015): 0.16 and 0.07.

Model validation

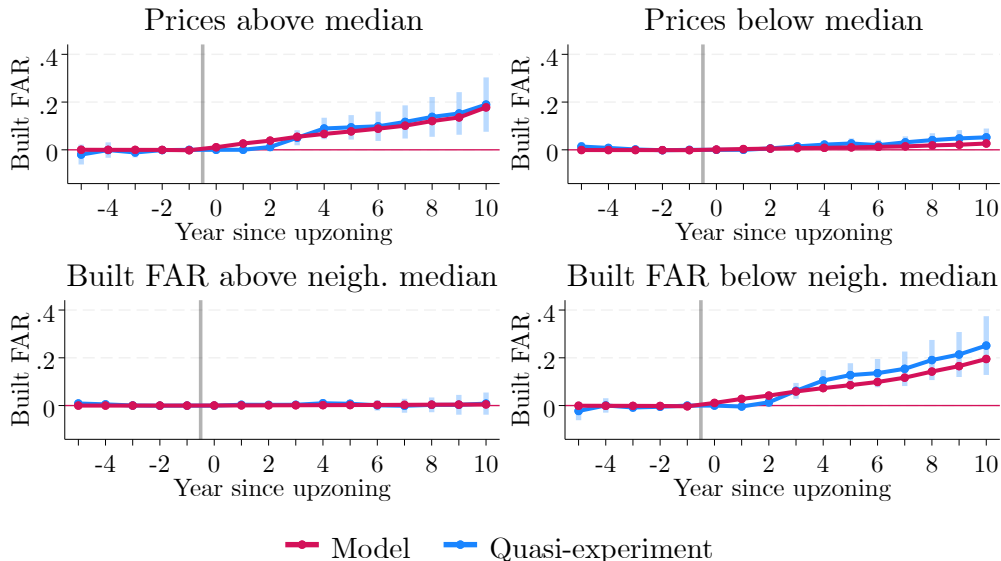
Predicted effect of upzonings

- Model estimation separate from reduced-form.
→ 90% of redevelopment on non-upzoned parcels.
- **Validation exercise:**
 - 1 Estimate model on non-upzoned parcels.
 - 2 Predict evolution of upzoned parcels with/without upzoning.
 - 3 Build model-implied effect of upzoning.
 - 4 Compare with quasi-experimental estimates.

Built FAR (including planned)



Model validation: Predicted effect of upzonings



Implications for zoning policy

Main counterfactuals

Simulate the evolution of the city until 2060, keeping fundamentals at their 2019 level.

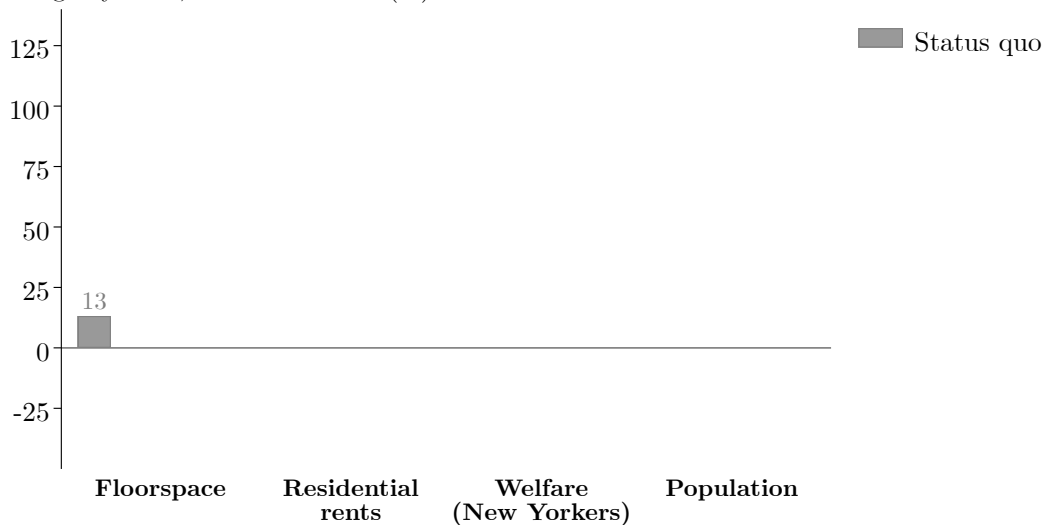
- ① **Status quo**: zoning stays as is.
- ② **Realistic but ambitious upzoning** around transit stations.
→ **Increases total allowed FAR in NYC by 60%.**
- ③ **No zoning** (excluding landmarks, historic districts, flood zones).

- ④ **Frictionless benchmark** (no zoning, no adjustment costs, price = marginal cost).

To what extent does zoning constrain
NYC's growth?

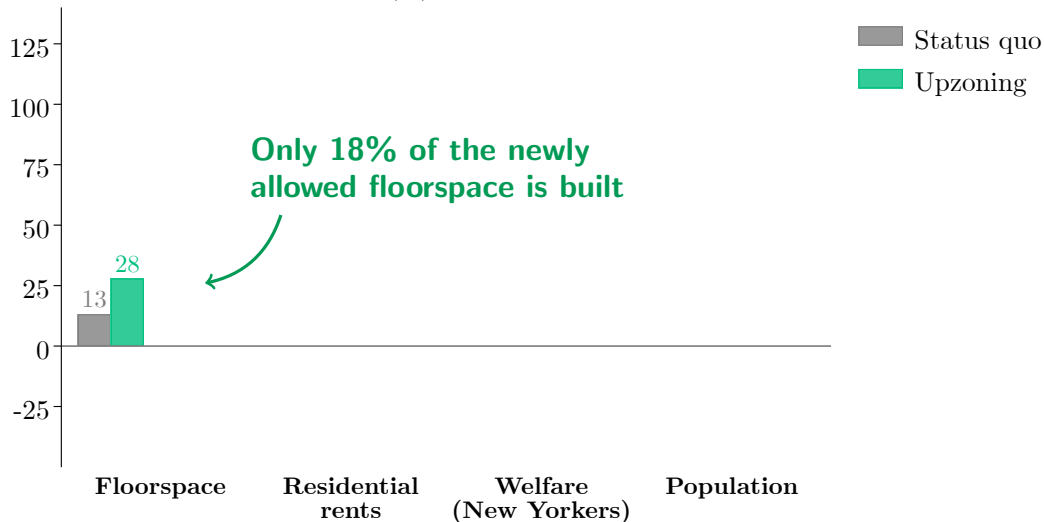
Under current zoning, NYC continues to grow slowly

Change by 2060, relative to 2019 (%)



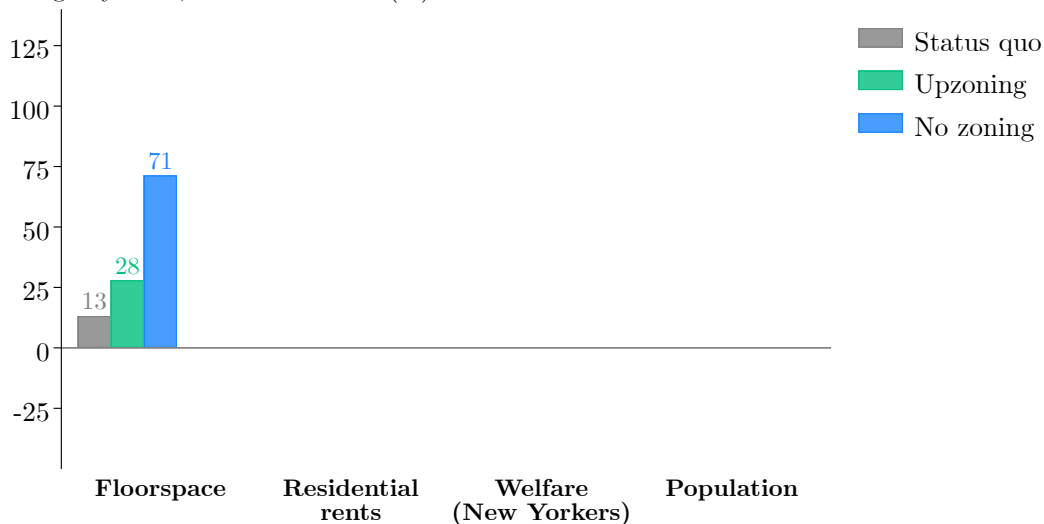
Upzoning increases floorspace by 15pp over 40 years

Change by 2060, relative to 2019 (%)



Completely removing zoning quintuples NYC's growth rate

Change by 2060, relative to 2019 (%)



Rent decreases are moderate

Change by 2060, relative to 2019 (%)



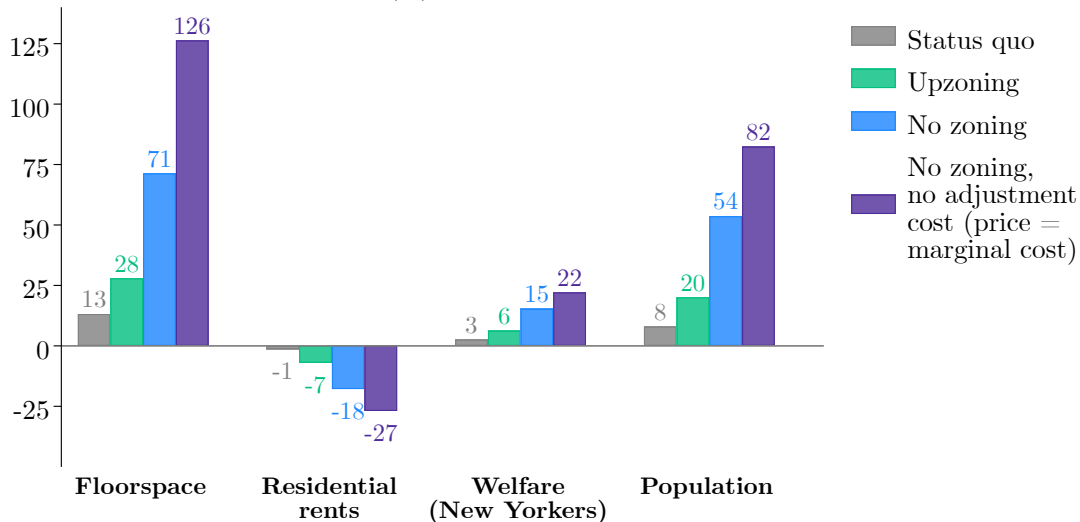
Rent decreases are moderated by migration

Change by 2060, relative to 2019 (%)

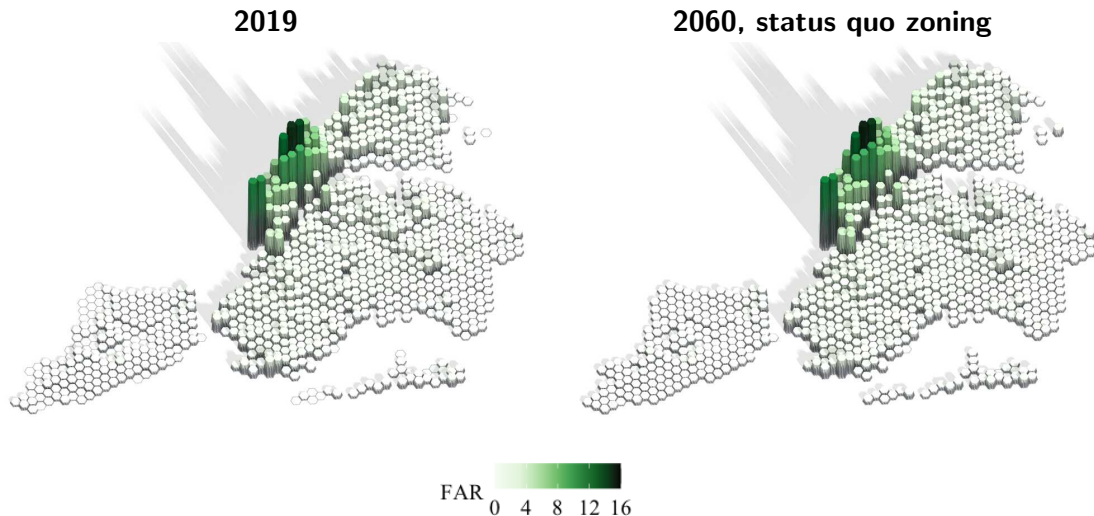


A model ignoring redevelopment greatly overstates effects

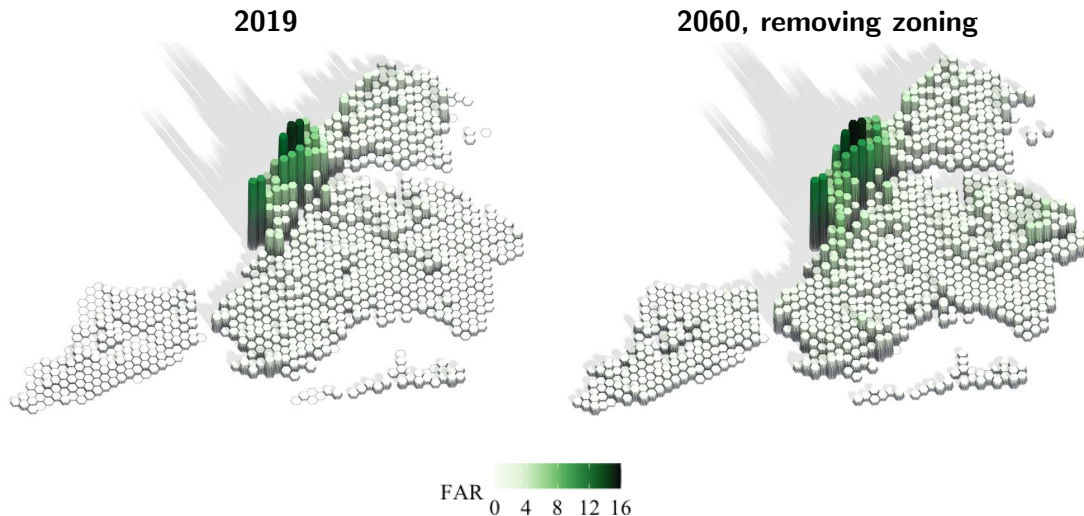
Change by 2060, relative to 2019 (%)



Technological constraints matter for credible counterfactuals



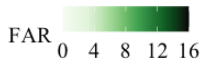
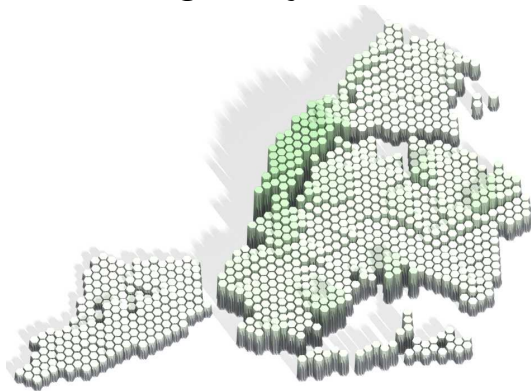
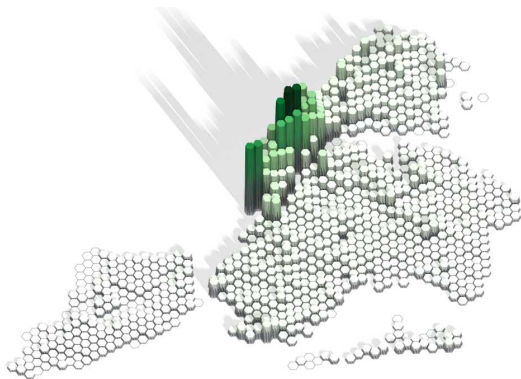
Technological constraints matter for credible counterfactuals



Technological constraints matter for credible counterfactuals

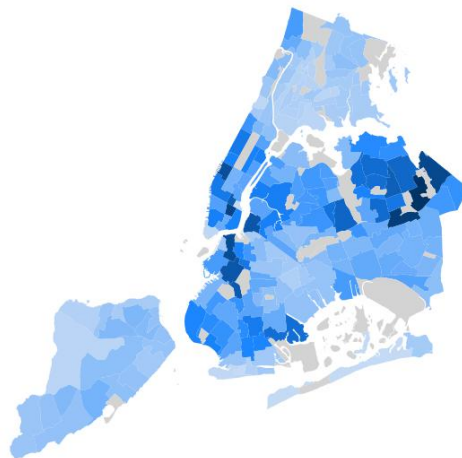
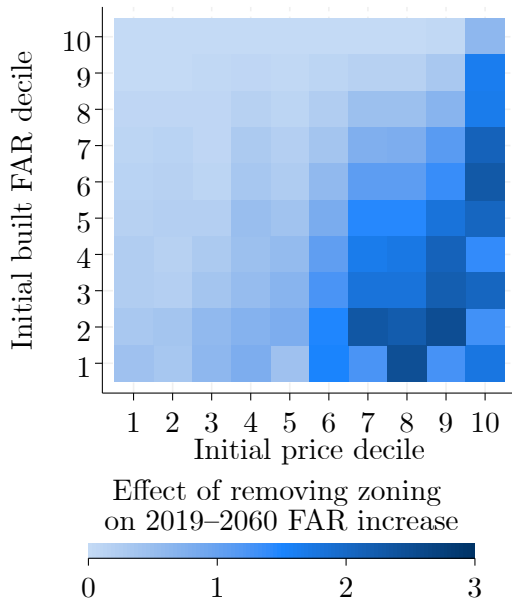
2019

No zoning, no adjustment costs

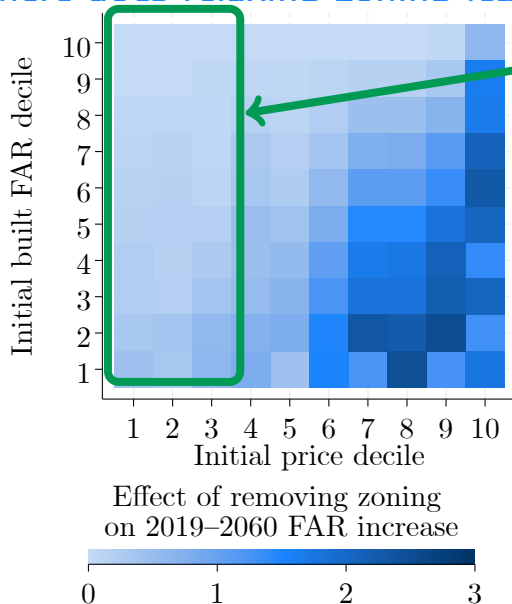


Where is zoning a constraint?

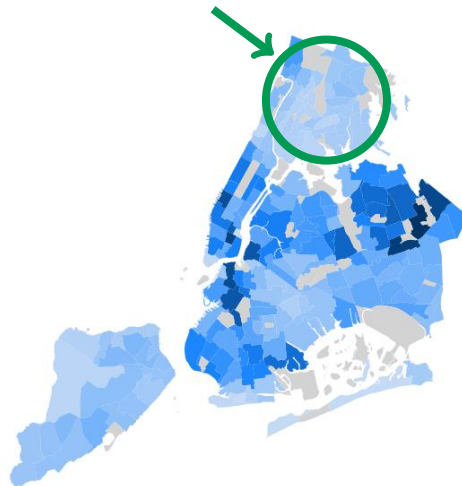
Where does relaxing zoning lead to increased supply?



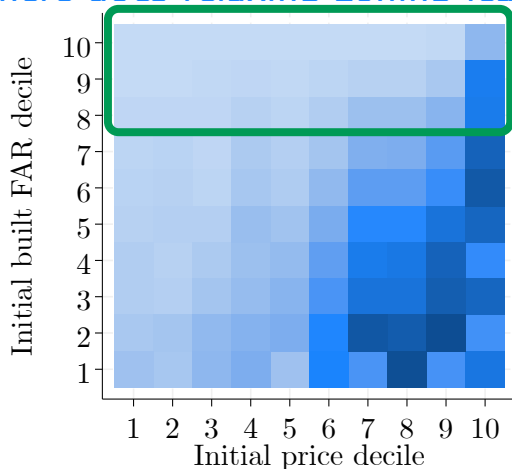
Where does relaxing zoning lead to increased supply?



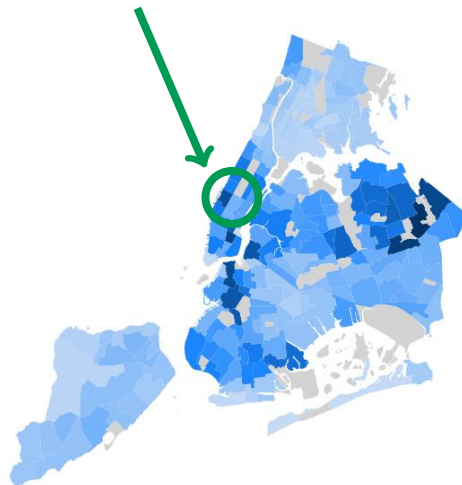
Zoning tends not to be binding when floorspace prices are low.



Where does relaxing zoning lead to increased supply?

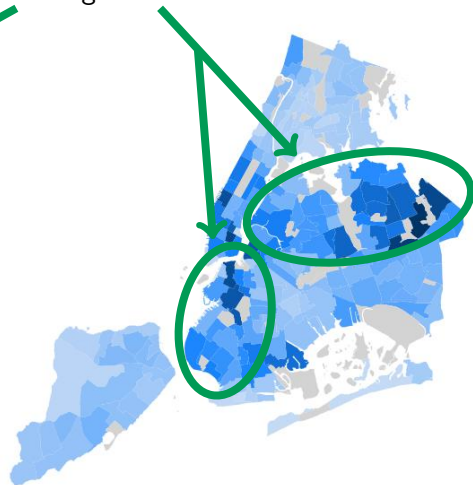
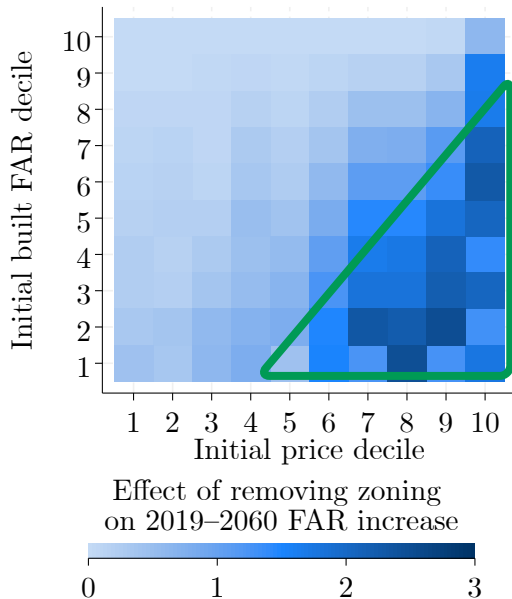


Effect of removing zoning
on 2019–2060 FAR increase

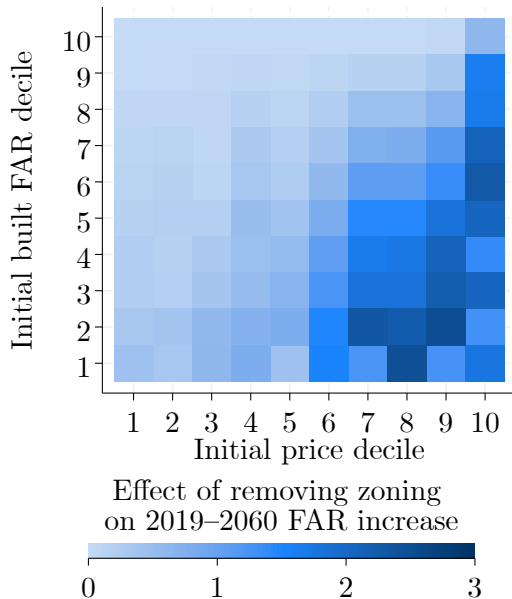


Where does relaxing zoning lead to increased supply?

Upzoning is effective when prices are high enough and density is low enough.



Where does relaxing zoning lead to increased supply?



- Upzonings should be **targeted**.
 - Upzoning is politically costly.
 - Can achieve half of the gains of removing zoning by deregulating 12% of the land.
- NYC's planners already do targeting.
- Upzoning would likely have limited effects in inexpensive cities.

Additional results

Additional results

- **Distributional effects.** [More](#)

- Lower-income workers benefit more.

- **Alternative policies** to promote construction and affordability. [More](#)

- Upzoning outperforms reducing construction costs and tax incentives.

- Contribution to the literature on the topic: Glaeser and Gyourko (2008), Soltas (2024)...

- **Supply elasticities.** [More](#)

- Compute supply elasticities at a granular level and different time horizons.

- Supply elasticities vary widely across neighborhoods. Zoning is their primary determinant.

- Contribution to the literature on the topic: Saiz (2010), Baum-Snow and Han (2024)...

Why did NYC's planners impose costly zoning regulations?

Why did NYC's planners impose such costly regulations?

- Zoning creates large welfare losses. Why is it in place?
- When the zoning code was crafted in 1961, restricting construction had limited costs.
 - Low population growth and land values, floorspace prices close to marginal cost.
 - Zoning could help curb negative spillovers from manufacturing.
- Much has changed since 1961:
 - Floorspace prices have skyrocketed, manufacturing activity has plummeted.
- But zoning has been much more persistent than planners anticipated and intended.
 - Rezoning is vulnerable to obstruction by those with a stake in the status quo.

Conclusion

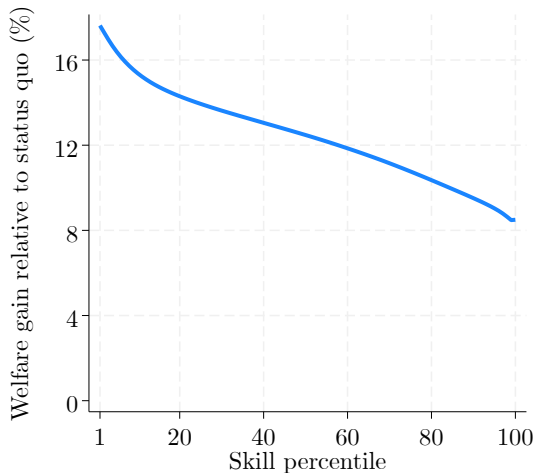
Conclusion

- **Redevelopment** is key to the economics of densely built areas.
 - Increases the supply of floorspace and reallocates land to most profitable use.
- This paper provides a framework to analyze this process.
 - I use it to study the effects of zoning.
 - Can be applied to other questions (rent control, transit upgrades, remote work, structural transformation, property taxation).
- Large fixed costs of redevelopment.
 - Even at a 40-year horizon, **take-up of upzoning is limited**.
 - Upzoning works best when **prices are high/density is low**.
- Gains **take time to materialize** and are **geographically spread out**.
 - Local reforms might look disappointing despite **large aggregate/long-term gains**.

Appendix

Who benefits from upzoning?

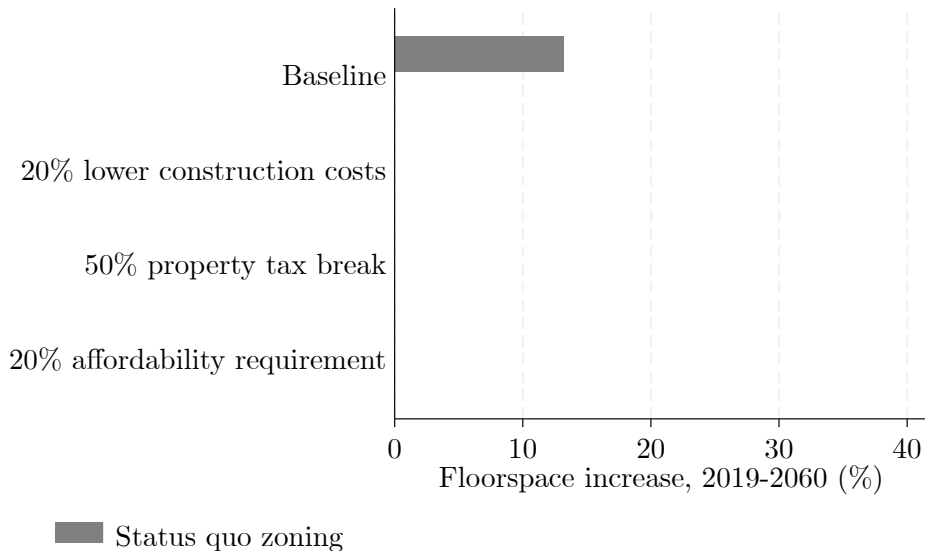
Removing zoning benefits low-income workers more [◀ Back](#)



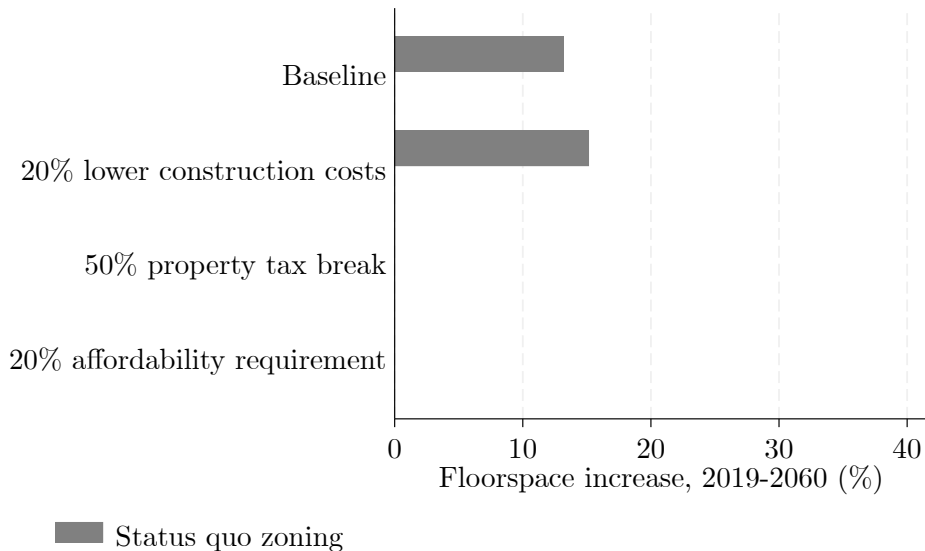
- Lower-income workers spend a larger share of income on housing.
- Increasing the housing stock lowers prices more in cheaper neighborhoods.

How effective are alternative policies to favor construction and affordability?

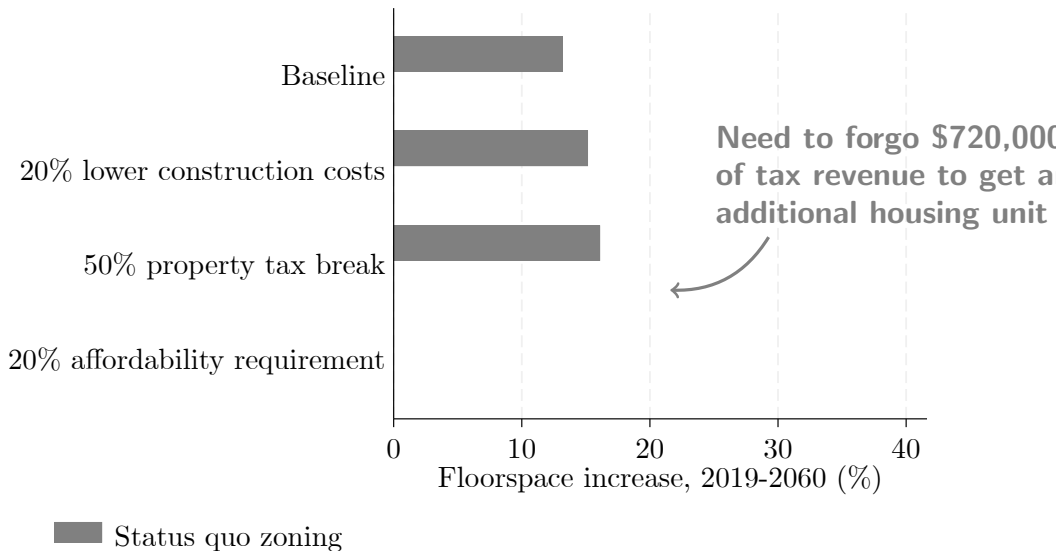
Alternative policies to favor construction and affordability



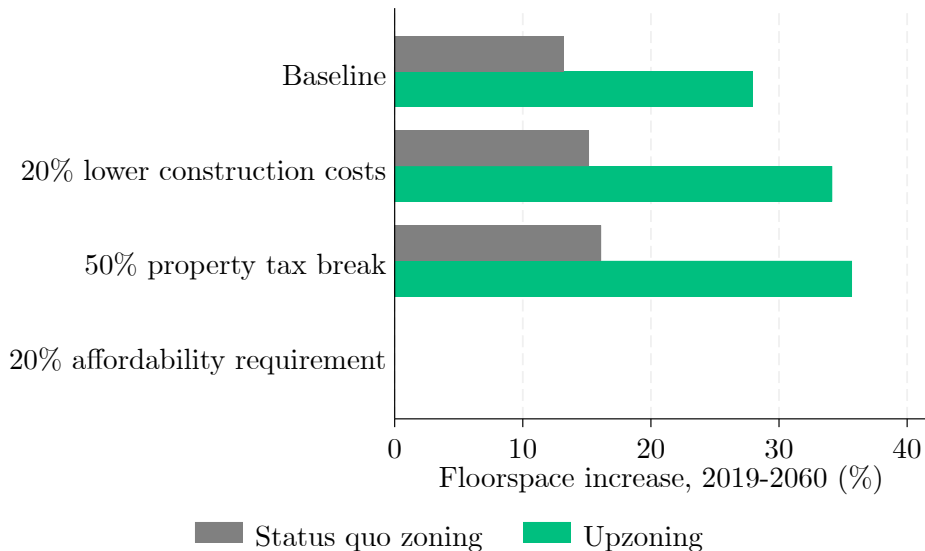
Lowering construction costs marginally boosts construction



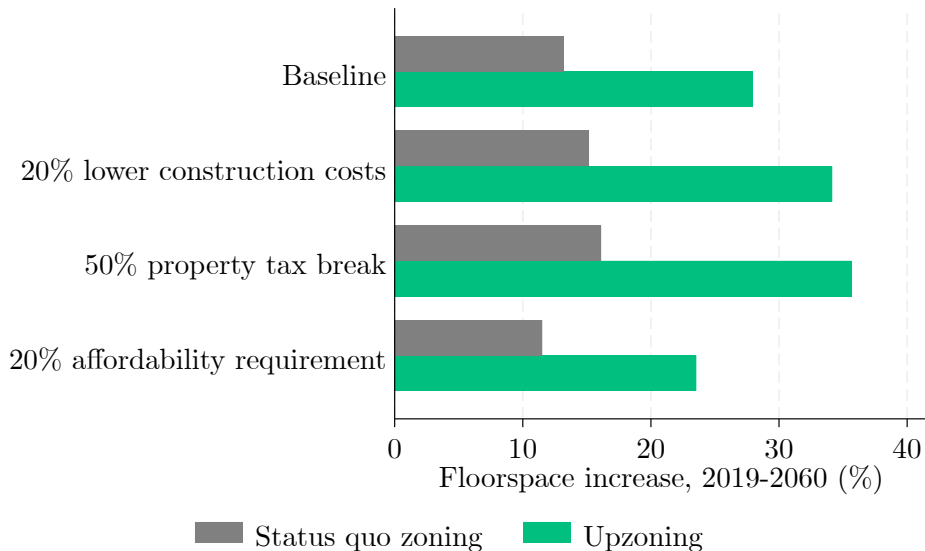
Tax breaks on new buildings are relatively ineffective and costly



Upzoning dominates alternative policies



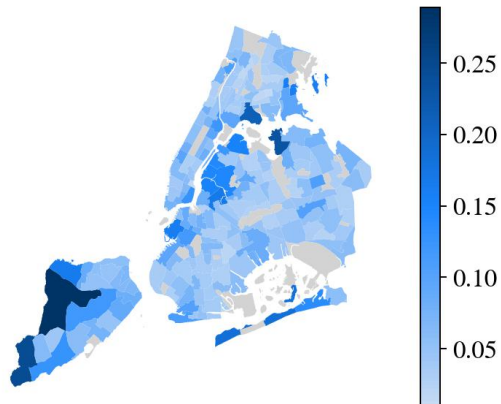
Inclusionary zoning: supply ↓ 0.6 sq.ft per affordable sq.ft

[◀ Back](#)

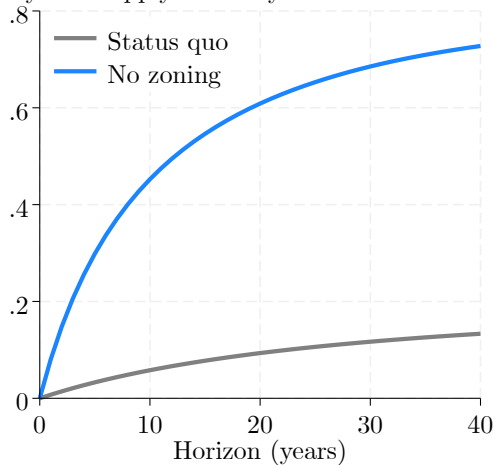
How does zoning affect floorspace supply elasticities?

Supply elasticities vary widely across neighborhoods

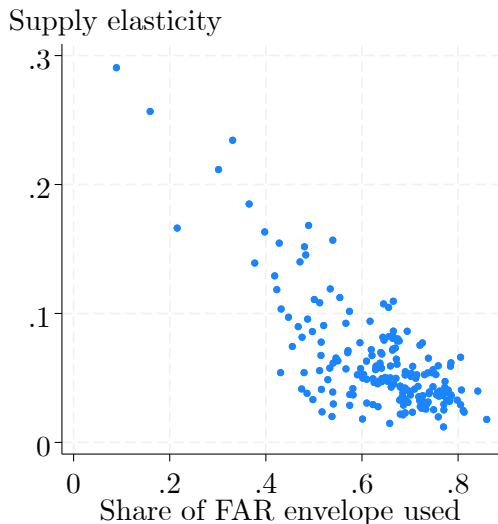
40-year supply elasticity
(status quo zoning)



Citywide supply elasticity



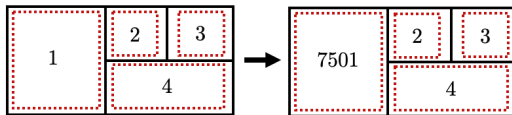
Zoning is the strongest correlate of supply elasticities [◀ Back](#)



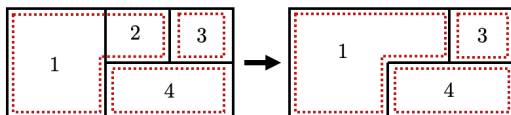
- Supply elasticities are largely determined by the share of the zoning envelope that has already been built out.
- Without zoning, citywide supply elasticity 5x higher.
 - Mostly determined by built density. [More](#)

Boundary changes (21,700 detected over 2004-2022)

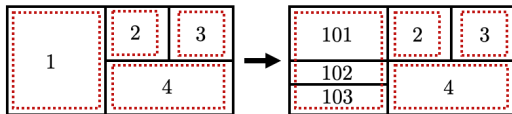
(a) Renaming



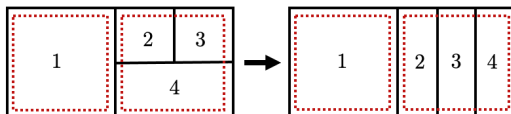
(b) Merge



(c) Split

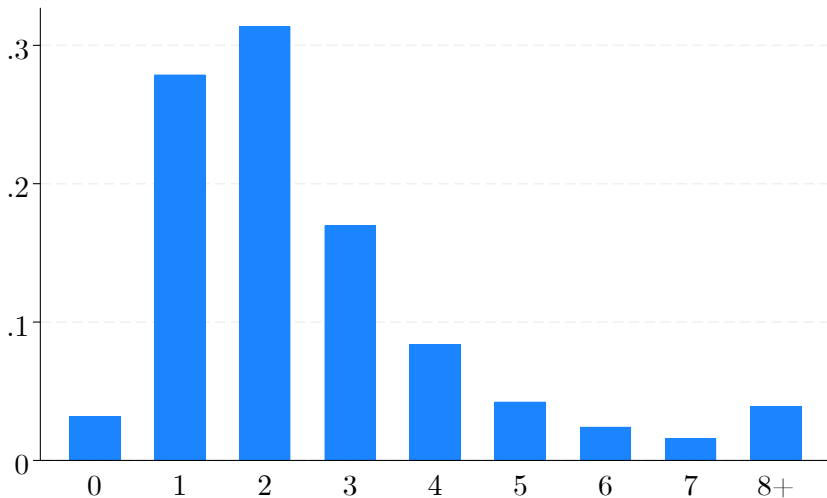


(d) Rearrangement



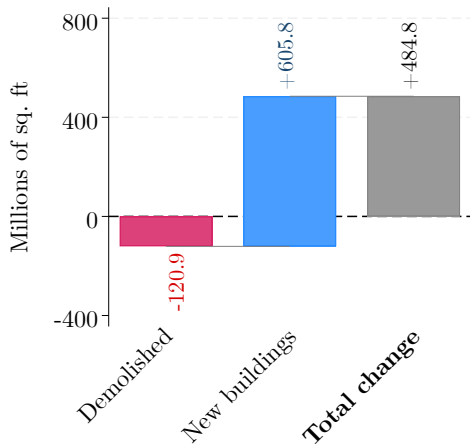
 NYC Zoning lot  Land parcel in the model

Redevelopment duration (in years)



Mature cities grow by redeveloping old structures

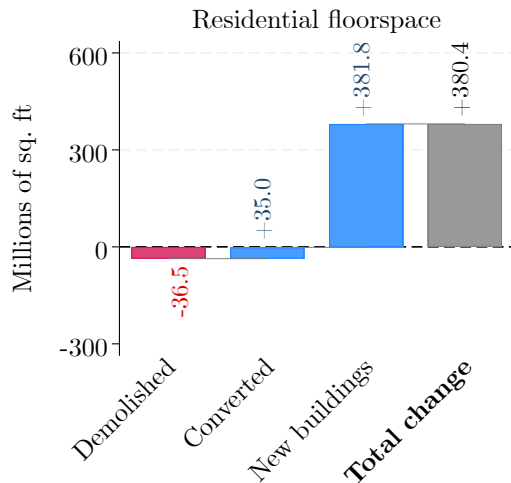
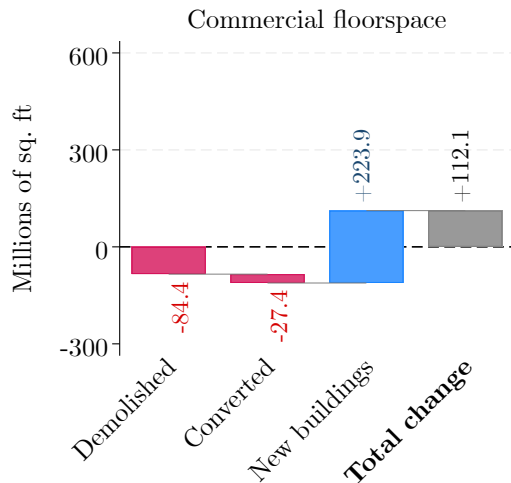
Evolution of total floorspace
(2004-2019)



- Since 2004, floorspace in NYC has grown at a rate of $\sim 0.6\%$ per year.

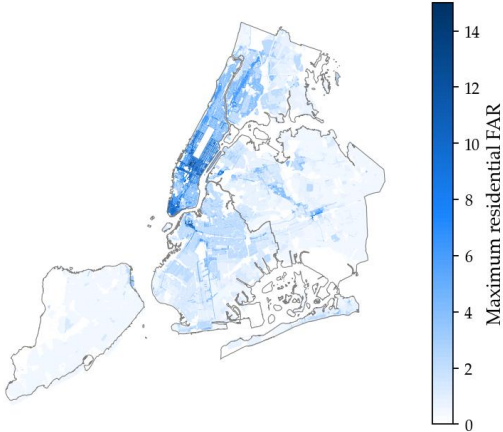
◀ Back

Changes in residential/commercial floorspace



FAR allowances in NYC's zoning map

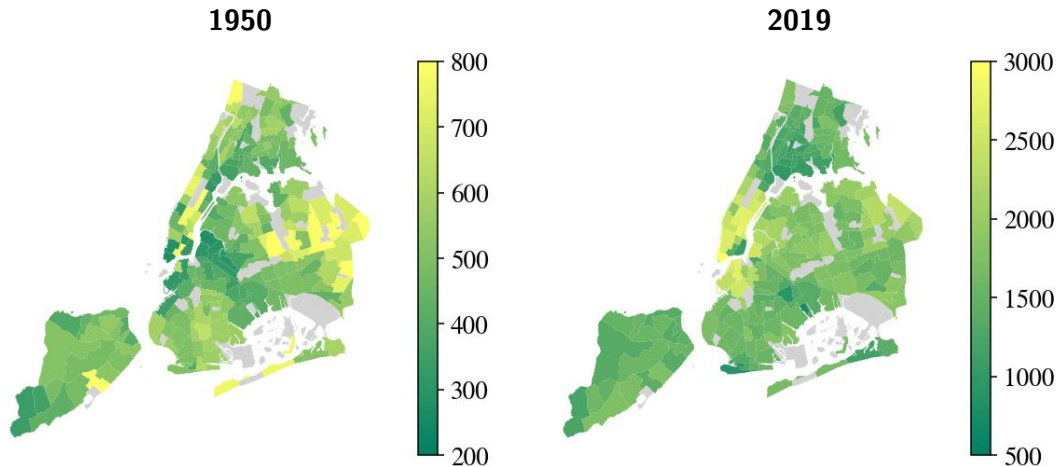
Residential uses



Commercial uses

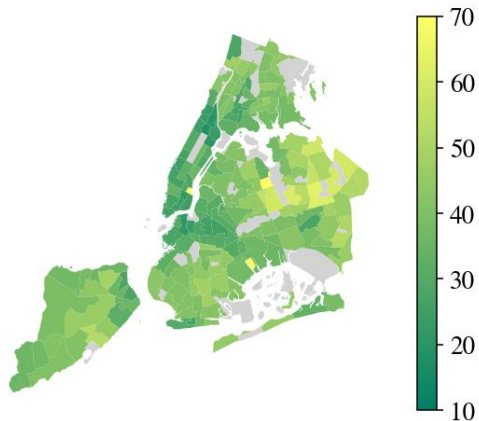


Evolution of median residential rents (\$/month)

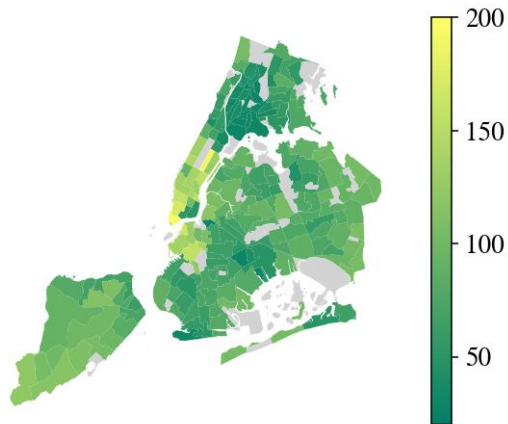


Evolution of median household income (\$k/year)

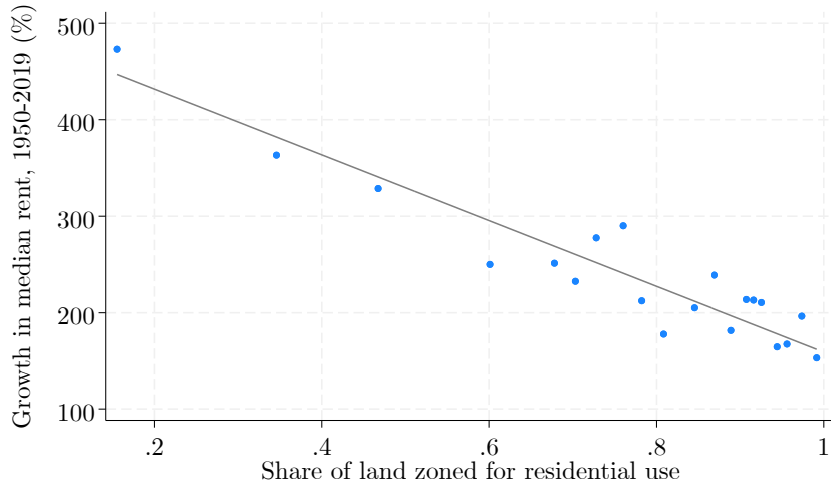
1950



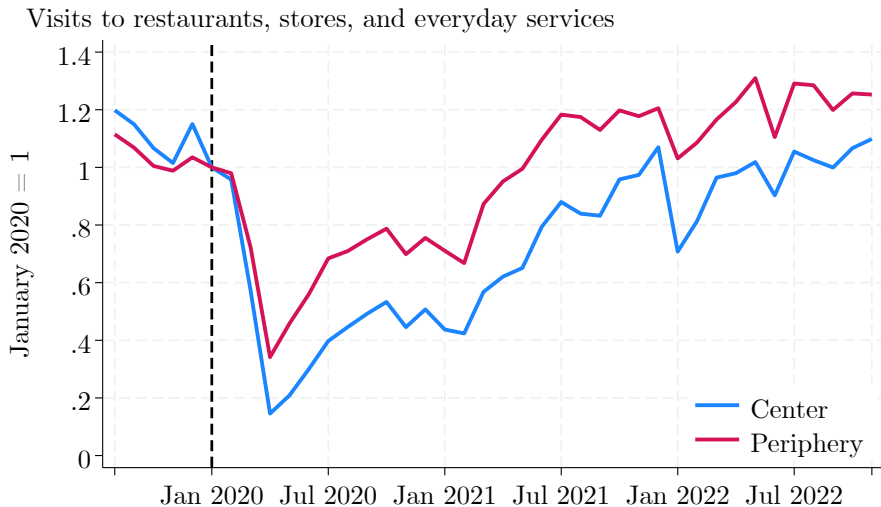
2019



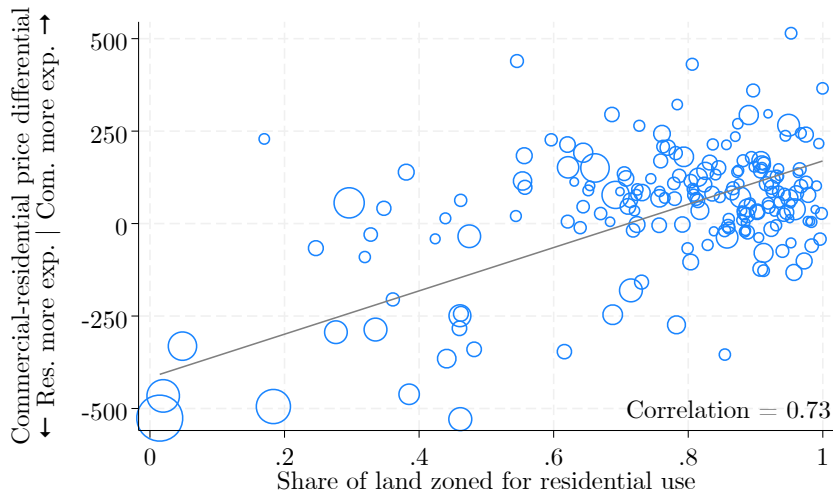
Commercial-oriented areas became more attractive over time



Work from home will likely accelerate existing trends



Use restrictions constrain the reallocation of land uses

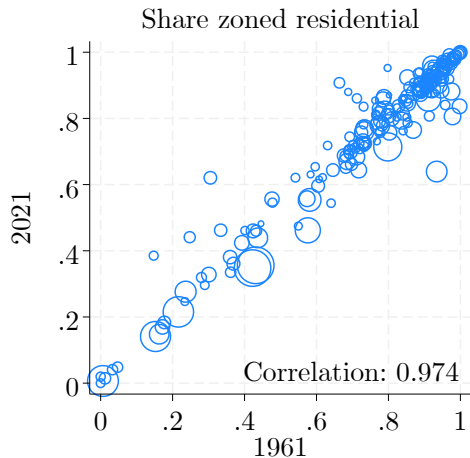
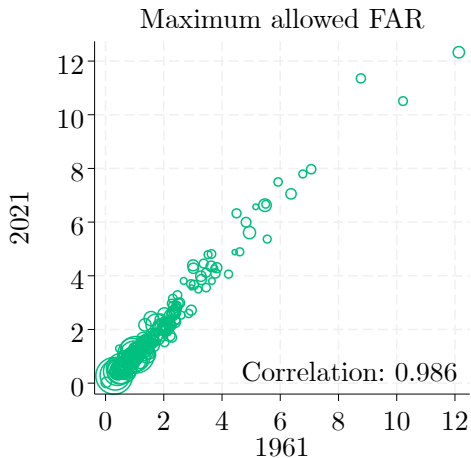


Zoning's goals

- Current zoning resolution adopted in 1961.
- Planners believed NYC has nearly reached its maximum size.
 - Didn't view the zoning code as restrictive.
- Aims of the 1961 zoning resolution:
 - 1 Promote tower-in-the-park development.
 - 2 Better separate commercial/residential uses.
 - De facto: stabilization of existing land uses.
- Current zoning regulations closely align with the 1961 ordinance.



Since 1961: Persistence in zoning

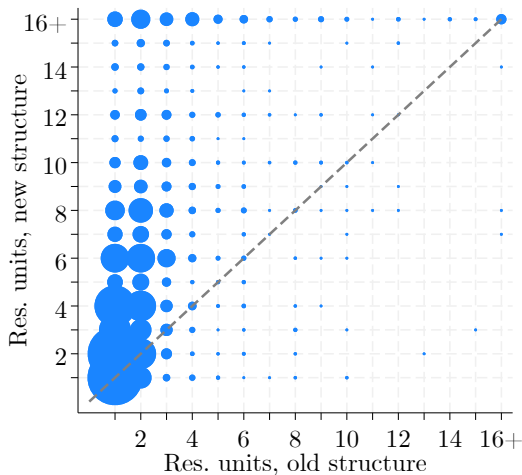


Change in number of units

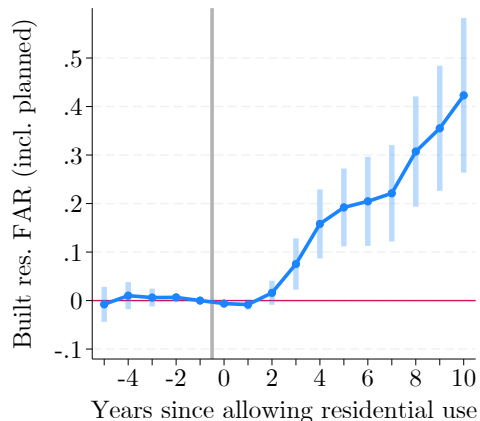
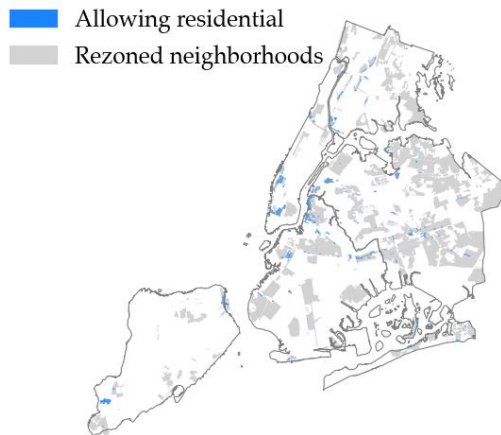
Residential-to-residential redevelopment

- New residential units are about 10% larger than the ones they replace.
- The number of units in new residential buildings is, on average, 3 times larger than in old structures.

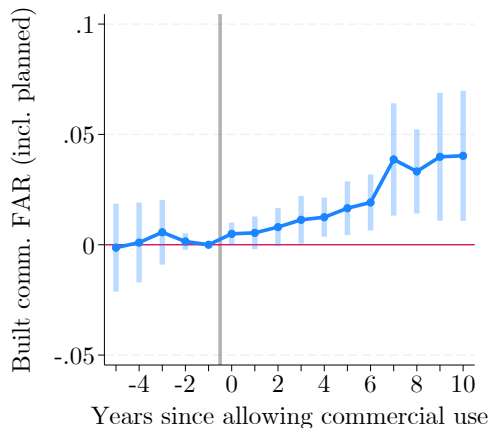
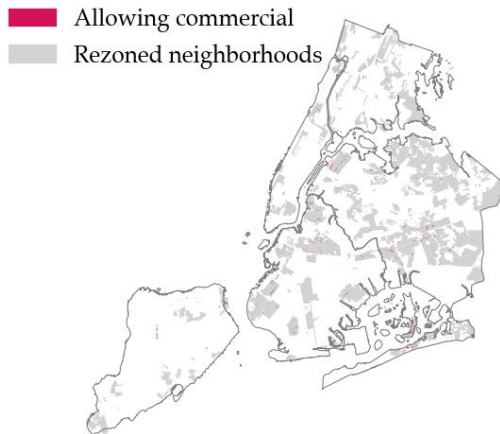
[◀ Back](#)



Effects of allowing residential use



Effects of allowing commercial use



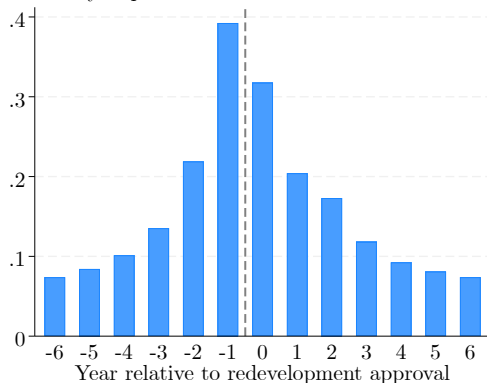
Probability of a parcel sale (probit regression)

	Probit coefficients	
Parcel sold		
Office space (% of total floorspace)	-0.076	(0.006)
Retail space (% of total floorspace)	-0.018	(0.004)
Garage space (% of total floorspace)	0.000	(0.007)
Storage space (% of total floorspace)	0.033	(0.009)
Factory space (% of total floorspace)	0.001	(0.007)
Hotel space (% of total floorspace)	-0.015	(0.020)
Other space (% of total floorspace)	-0.327	(0.006)
Condo/Coop	-0.684	(0.008)
Parcel in Bronx	0.051	(0.003)
Parcel in Brooklyn	0.038	(0.003)
Parcel in Queens	0.036	(0.003)
Parcel in Staten Island	0.023	(0.003)
Constant	-1.615	(0.003)
Observations	13,156,064	

Parcel sales around redevelopment events

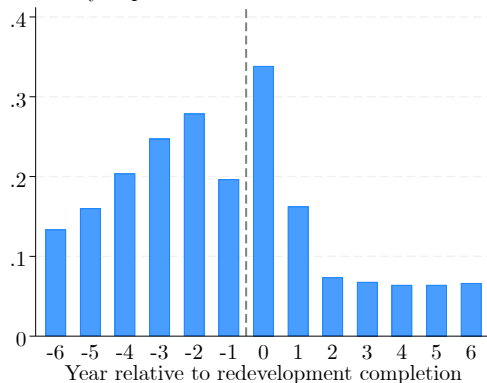
Around project approval

Probability of parcel transaction



Around project completion

Probability of parcel transaction



Estimation details: Building values

- Using the sales data, I estimate separately for residential and commercial structures:

$$\log(\text{price}_s) = \underbrace{\rho_{1,n(s)}^\theta}_{\text{Neigh. FE}} + \underbrace{\rho_{2,bt(s)}^\theta}_{\text{Borough} \times \text{year FE}} + \beta \underbrace{(\mathbf{x}_s - \bar{\mathbf{x}}_s)}_{\text{Structure characteristics}} + \nu_s$$

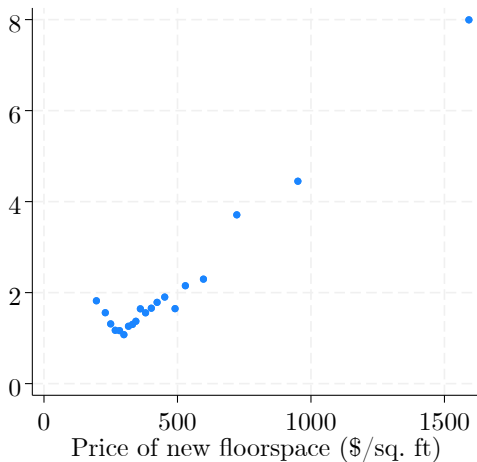
- Expected value of floorspace in a new building: $\bar{p}_{nt} = \exp(\hat{\rho}_{1,n} + \hat{\rho}_{2,b(n)t} + \hat{\sigma}_\nu/2)$
- Negative coefficient on (log) FAR, more negative for commercial.
 - Less usable floorspace in tall buildings (e.g., because of mechanical space).
 - Lower floor of commercial buildings is more valuable.

Hedonic regression coefficients [◀ Back](#)

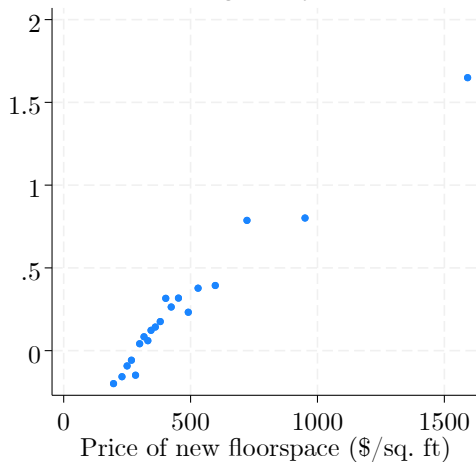
	(1)		(2)	
	Residential		Commercial	
(log) Built FAR	-0.027	(0.002)	-0.091	(0.011)
(log) Unit size	-0.065	(0.002)	-0.108	(0.005)
Age	-0.002	(0.000)	0.000	(0.000)
Rent-stabilized	-0.336	(0.002)		
Landmark	-0.078	(0.006)	0.087	(0.060)
Grade A	0.162	(0.003)	0.222	(0.032)
Grade B	0.025	(0.002)	0.095	(0.019)
Grade C	0.007	(0.002)	-0.024	(0.019)
Brick	-0.155	(0.003)	-0.001	(0.084)
Frame	-0.068	(0.003)	-0.224	(0.118)
Masonry	-0.156	(0.003)	-0.071	(0.019)
Office building			0.147	(0.025)
Retail building			0.222	(0.021)
Garage building			0.004	(0.025)
Industrial building			0.030	(0.025)
Hotel			0.352	(0.048)
Neighborhood FE	Yes		Yes	
Borough × Year FE	Yes		Yes	
Observations	428,338		15,795	

Developers build taller when facing high prices

Built FAR



FAR points above regulatory limit



Estimated cost parameters

(a) Variable cost parameters

α^0	Baseline cost of materials	80.4	(2.3)
ζ	Capital cost share	0.51	(0.005)
σ_η	Cost shock standard deviation	1.08	(0.02)

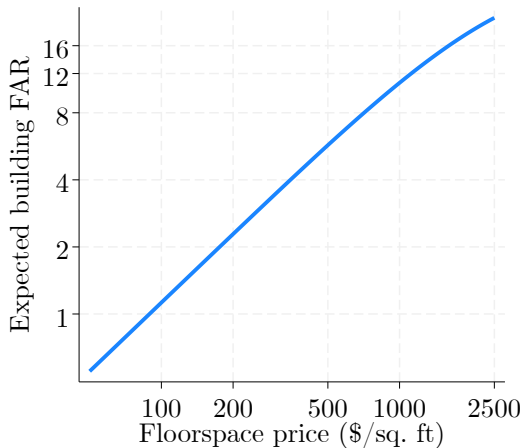
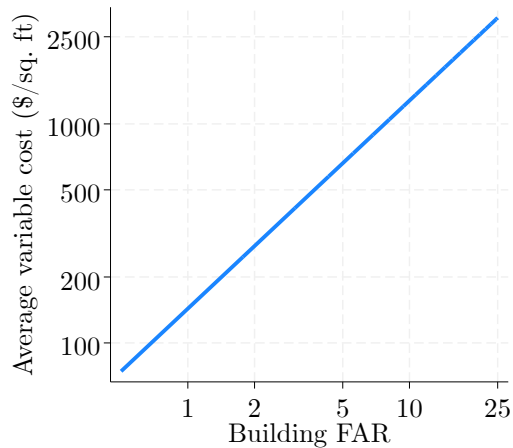
(b) Fixed cost parameters

δ^0	Base fixed cost	175.2	(7.7)
$\delta^{\text{demolition}}$	Demolition multiplier	1853.8	(37.6)
δ^{density}	Neighborhood density multiplier	318.5	(11.3)
$\delta^{\text{protected}}$	Protected parcels multiplier	0.77	(0.03)

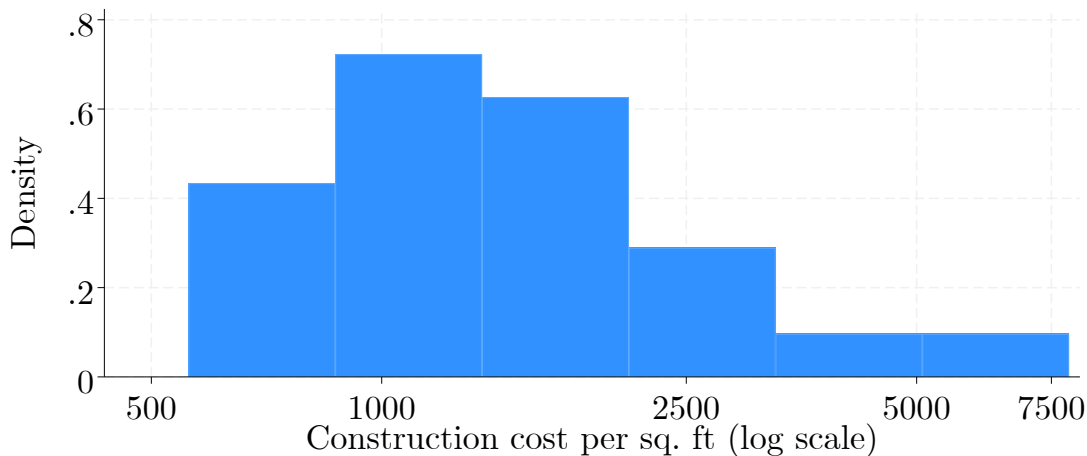
(c) Profit shock parameters

σ_ϵ^0	Base profit shock variance	73.7	(2.9)
$\sigma_\epsilon^{\text{demolition}}$	Demolition multiplier	138.8	(5.1)
$\sigma_\epsilon^{\text{density}}$	Neighborhood density multiplier	248.2	(4.9)

Construction costs



Construction costs for skyscrapers

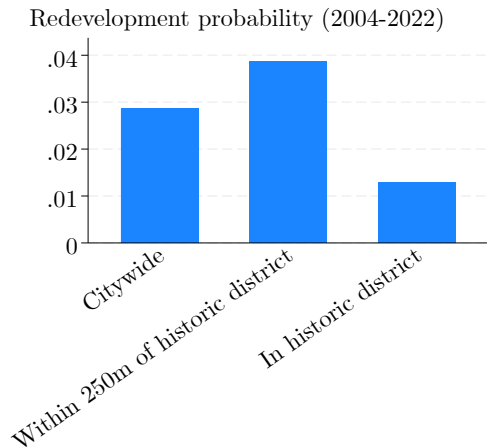


Fixed costs parameterization

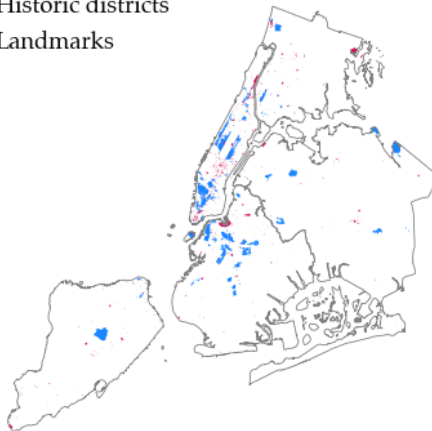
$$FC_{it} = (\delta^0 + \delta^{\text{demolition}} h_{it}^{\text{old}} + \delta^{\text{density}} \bar{h}_{n(i)t}) (1 + \delta^{\text{protected}} \mathbf{1}_{it}^{\text{protected}}).$$

- δ^0 : Baseline fixed cost.
- h_{it}^{old} : FAR of building to demolish.
- $\bar{h}_{n(i)t}$: Average neighborhood density.
- $\mathbf{1}_{it}^{\text{protected}}$: Landmarked parcel or in historic district.

Protected areas



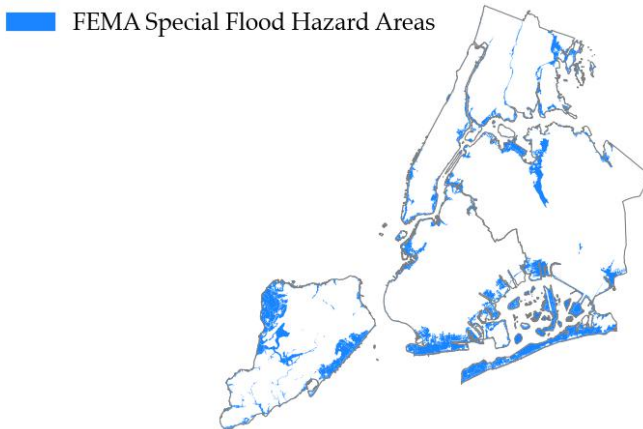
Historic districts
Landmarks



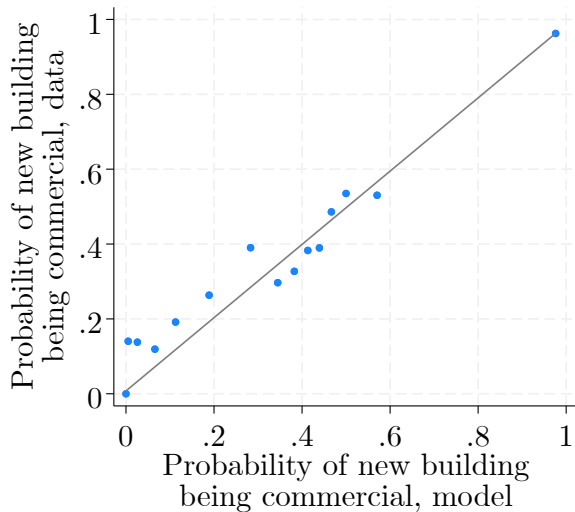
◀ Back (fixed costs)

◀ Back (counterfactuals)

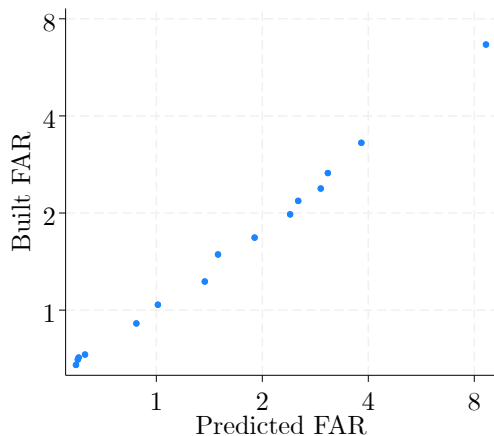
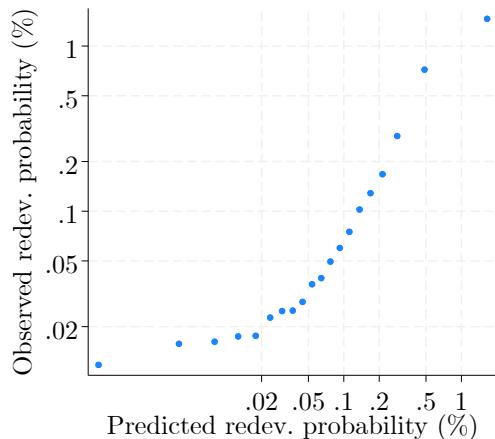
Flood zones

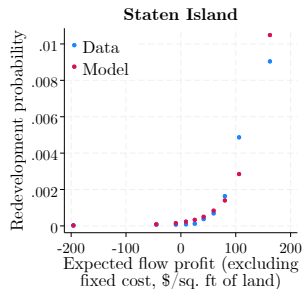
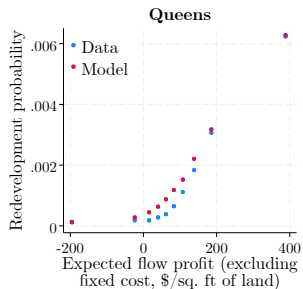
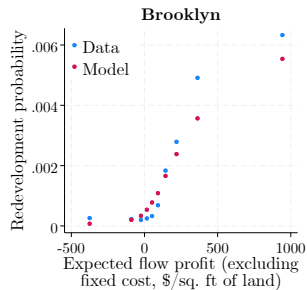
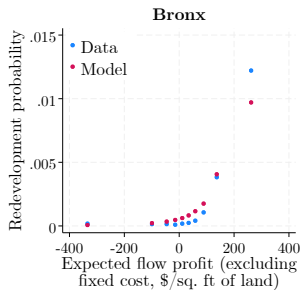
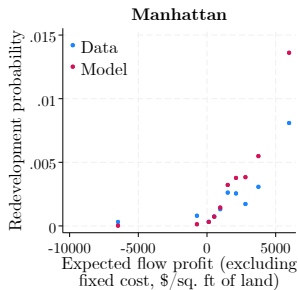


Model fit: Residential vs. commercial construction

[◀ Back](#)

Fit on extensive and intensive margins





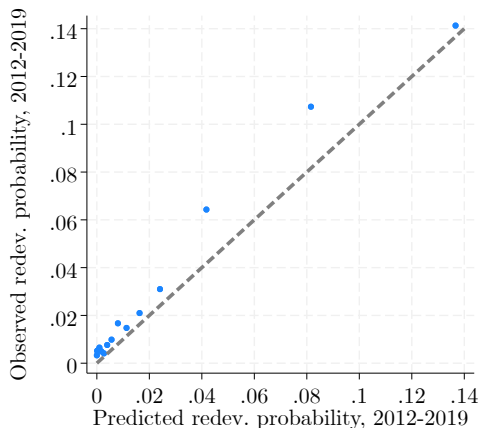
Out-of-sample model fit [◀ Back](#)

- Re-estimate the model using first half of the data (2004–2011):
 - Price levels;
 - Preferences and location fundamentals;
 - Sale probabilities;
 - Variable costs;
 - Fixed costs.
- Predict the evolution of the city over 2012–2019.

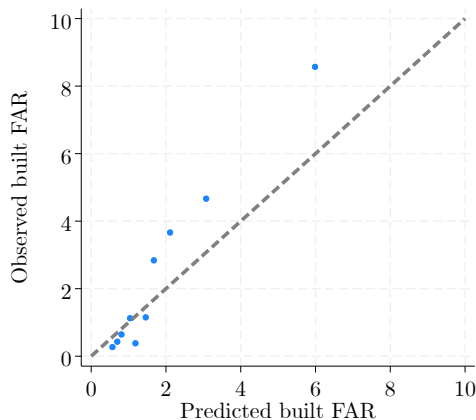
Out-of-sample model fit: Parcel level

[◀ Back](#)

Redevelopment probabilities



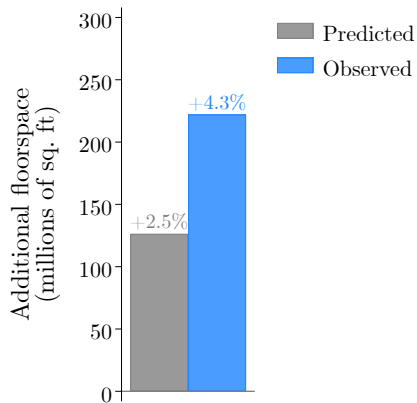
Built FAR of redeveloped buildings



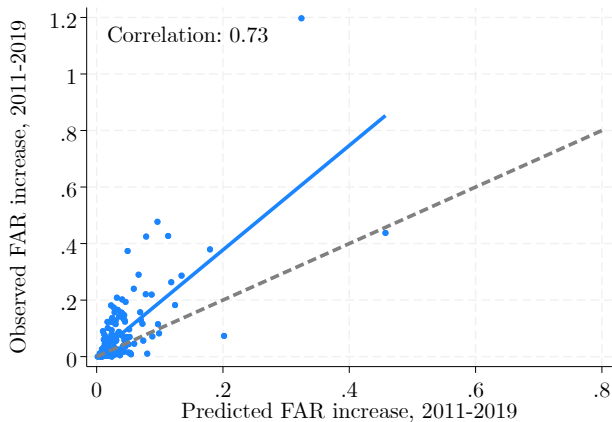
→ Differences explained by high price growth post-2012.

Out-of-sample model fit: Aggregate growth [◀ Back](#)

Citywide floorspace growth



Neighborhood-level growth



→ Differences explained by high price growth post-2012.

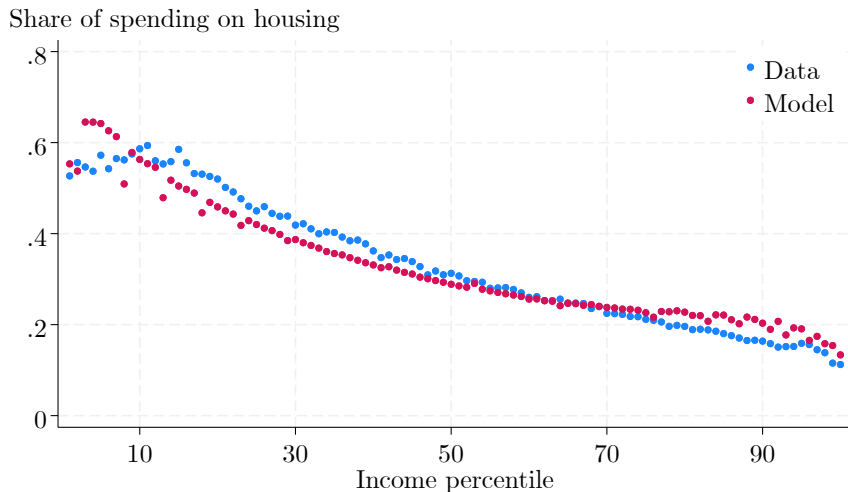
Demand model estimation (1)

- Calibrate β to 0.1 using the ACS.
- Estimate the shape of z^W at 4.4 using the commuting data.
- Calibrate wages from the number of people working in each location.
 - High-wage locations attract more workers, and from further away.
- Calibrate amenities $\mathbf{B}$ with residential prices and the number of residents in each location.
 - Locations attracting many residents despite high prices must have high amenities.
- Calibrate subsistence levels $\bar{h}$ with the total housing consumption in each location.
 - Higher levels of $\bar{h}$ lead to more housing consumption.
 - Average $\bar{h}$ of 224 sq. ft (IQR = [170 sq. ft, 265 sq. ft])

Demand model estimation (2)

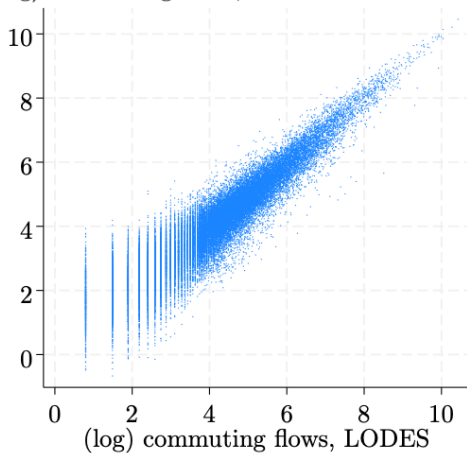
- Calibrate the shape of z^H to 2.9 match the variance of average neighborhood incomes.
 - Higher variance of $z^H \implies$ less sorting across neighborhood by income.
- Calibrate productivities $\mathbf{A}$ and floorspace shares in production α using data on commercial floorspace quantities and prices, the number of jobs in each location, and the calibrated wages.
 - Productivity is estimated to be higher near the center of the city.
 - α averages 0.18 across neighborhoods (IQR = [0.14, 0.21]). Close to benchmarks in the literature (0.2 in Ahlfeldt et al., 2015; 0.16 in Greenwood, Hercowitz, and Krusell, 1997).

Engel curve for housing

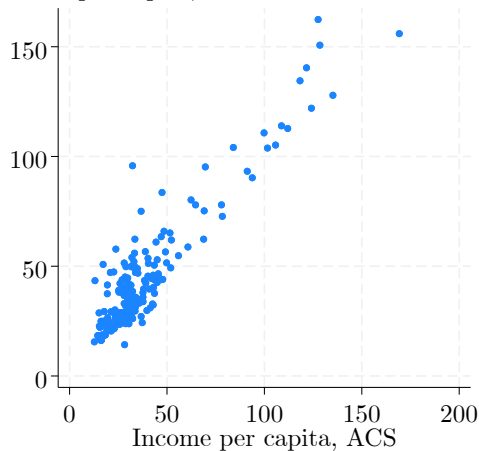


Demand model fit: Commuting flows and sorting

(log) commuting flows, model

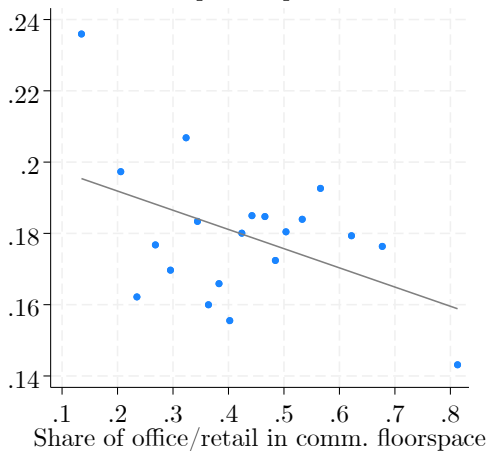


Income per capita, model

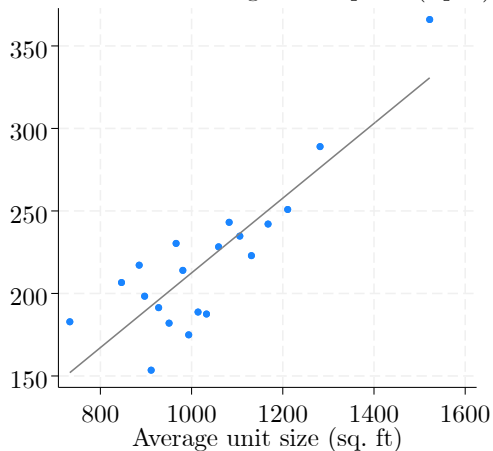


Demand model fit: α and $\bar{h}$

Est. share of floorspace in production



Estimated min. housing consumption (sq. ft)



Amenity	Correlation between share of satisfied residents and <i>B</i>
Neighborhood cleanliness	0.47
Control of street noise	0.07
Household garbage pick-up	0.33
Recycling services	0.34
Snow removal	0.77
Rat control	0.36
Bike safety	0.25
Pedestrian safety	0.53
Street maintenance	0.53
Parking enforcement	0.74
Storm water drainage and sewer maintenance	0.57
Availability of healthcare services	0.52
Availability of cultural activities	0.60
Neighborhood parks	0.86
Fire protection services	0.75
Emergency medical services	0.73
Neighborhood public safety	0.66
Bus services	0.66
Subway services	0.71
Public services	0.63

Large construction events

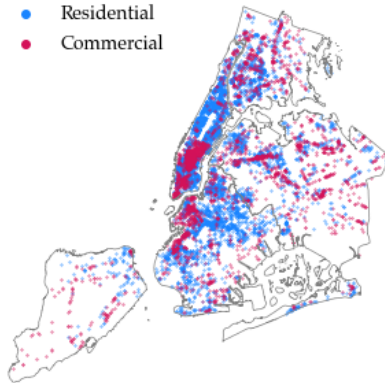
Construction events

- Residential
- Commercial

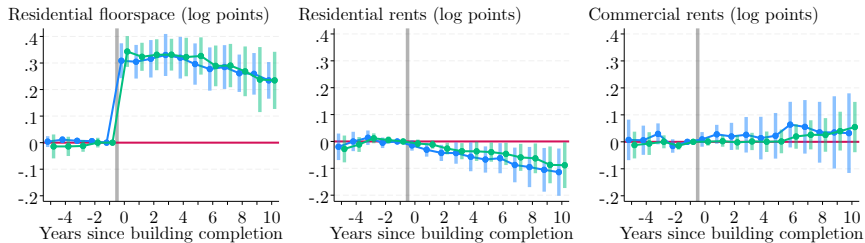


Buildings with rent data

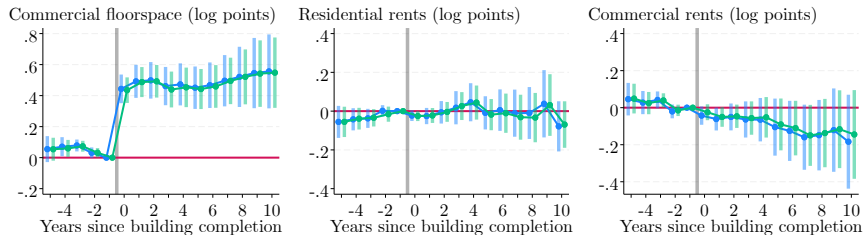
- Residential
- Commercial



Effects of residential construction

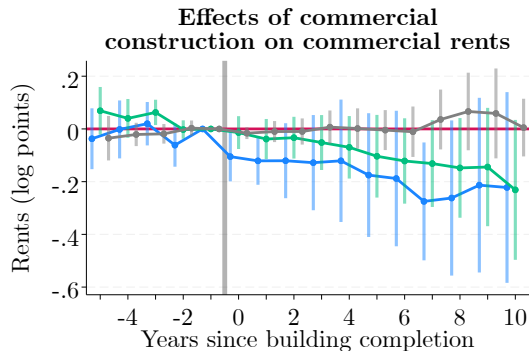
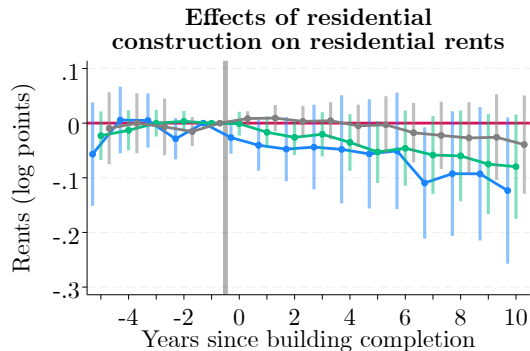


Effects of commercial construction



● Baseline ● With overlapping events

Spatial decay of price effects



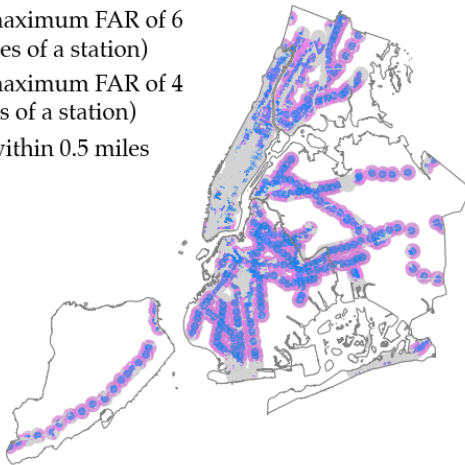
—●— 0-250 ft —●— 250-500 ft —●— 500-750 ft

Estimated spillover parameters

Parameter	Interpretation	Calibrated value	Targeted elasticity
γ^{RR}	Effect of residents on amenities	0.11	$\varepsilon^{RR} = -0.42$
γ^{RC}	Effect of residents on productivity	0.03	$\varepsilon^{RC} = 0.14$
γ^{CR}	Effect of jobs on amenities	-0.03	$\varepsilon^{CR} = -0.03$
γ^{CC}	Effect of jobs on productivity	0.07	$\varepsilon^{CC} = -0.28$

Transit-Oriented Development

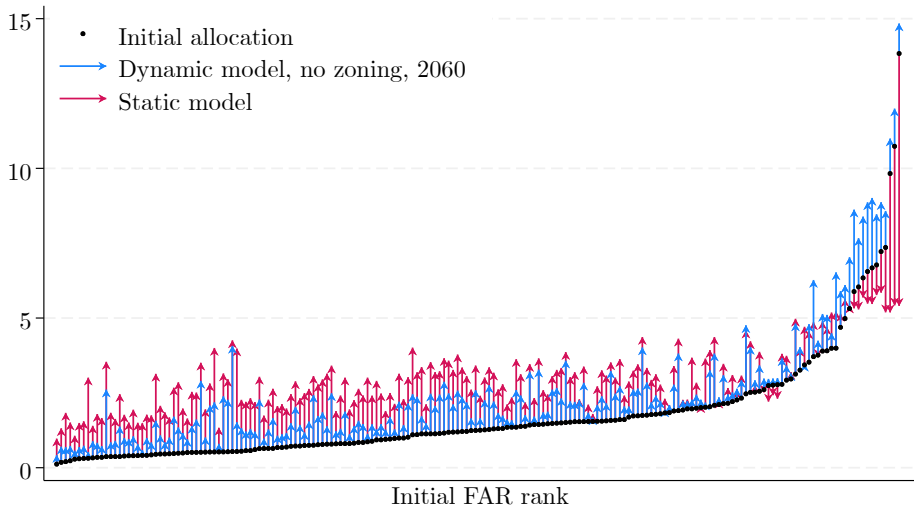
- Upzoned to a maximum FAR of 6 (within 0.25 miles of a station)
- Upzoned to a maximum FAR of 4 (within 0.5 miles of a station)
- Not upzoned, within 0.5 miles of a station



→ Upzones 37% of the city, mostly in outer boroughs.

Dynamic vs. static model

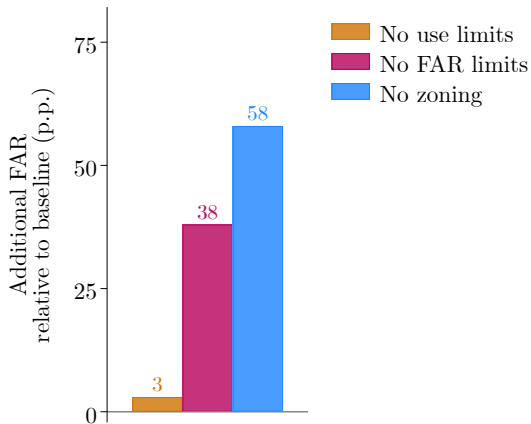
Built FAR



FAR vs. use limits [◀ Back](#)

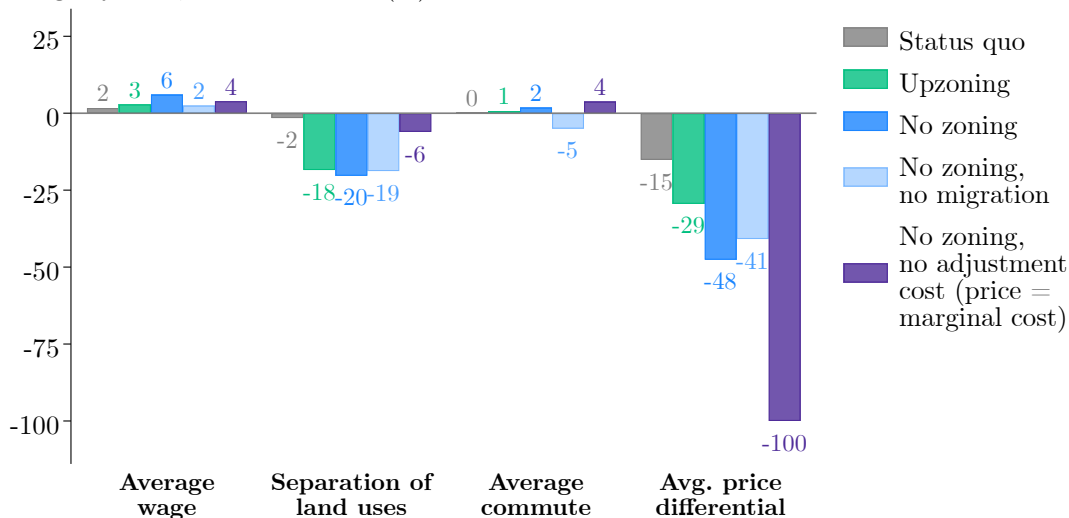
- Relaxing use limits does not spur much redevelopment.
 - Need to relax FAR limits to make redevelopment profitable.
- FAR and use limits interact.

Effects on floorspace over 40 years

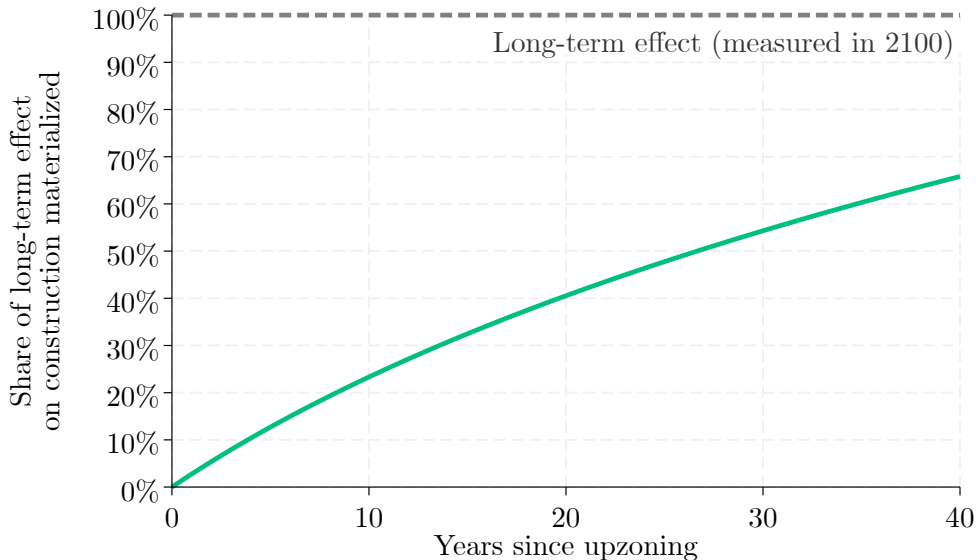


Additional outcomes

Change by 2060, relative to 2019 (%)



Effects of upzoning materialize slowly over time [◀ Back](#)



Use vs. FAR limits

Change by 2060, relative to 2019 (%)



Use vs. FAR limits

Change by 2060, relative to 2019 (%)



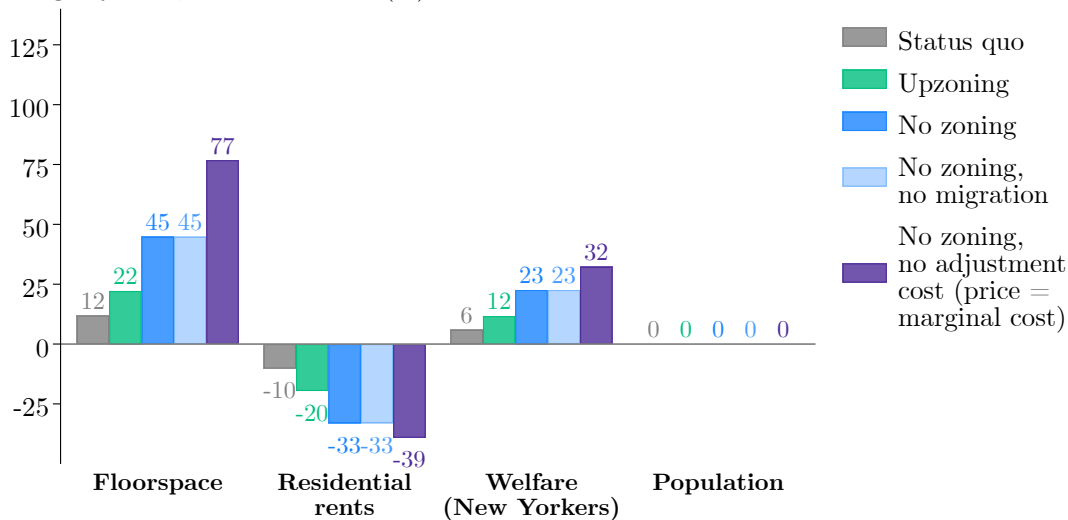
No spillovers

Change by 2060, relative to 2019 (%)

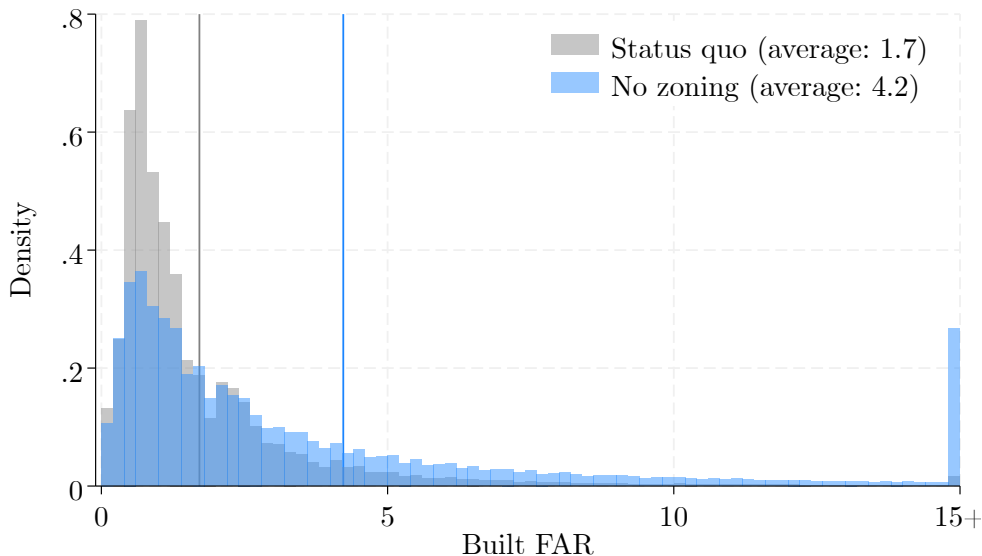


No migration

Change by 2060, relative to 2019 (%)



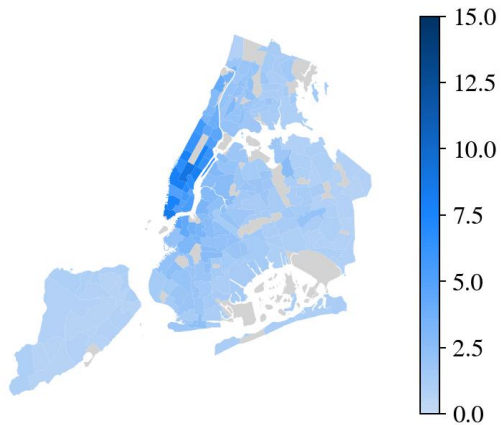
FAR distribution of new buildings

[◀ Back](#)

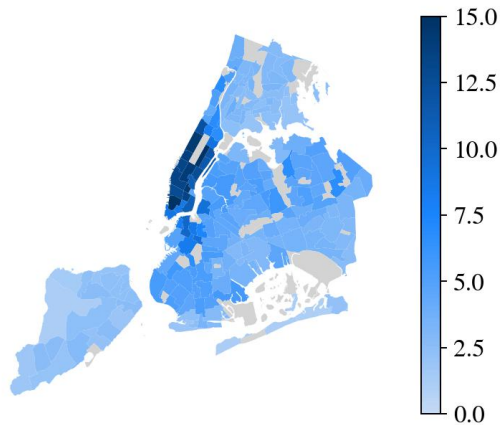
Average FAR of new buildings

[◀ Back](#)

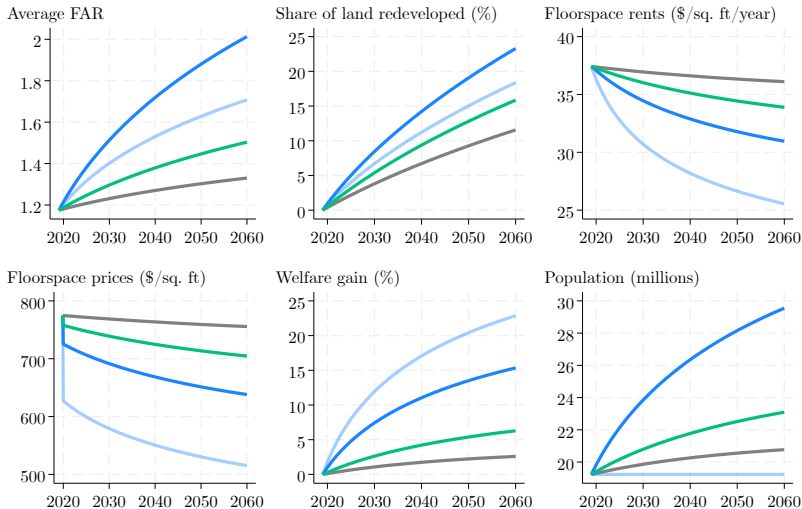
(a) Status quo



(b) No zoning



Transition paths

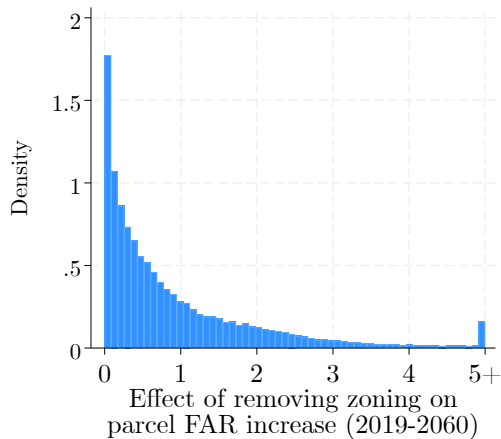
[◀ Back](#)

— Status quo — No zoning
— No zoning, no migration — Upzoning

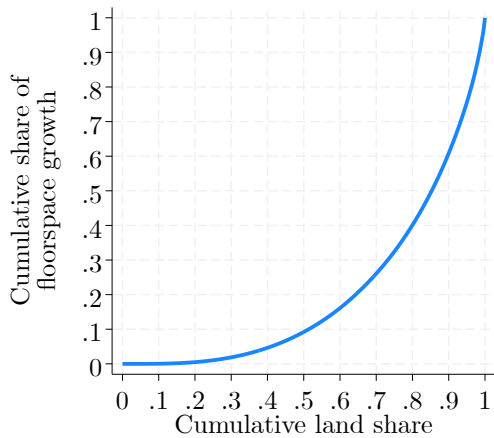
Upzonings' effects are concentrated

[◀ Back](#)

**Parcel-level distribution
of FAR increases**

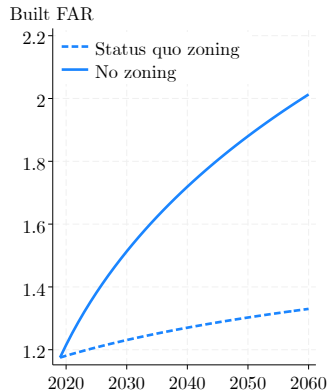


**Spatial concentration
of floorspace growth**

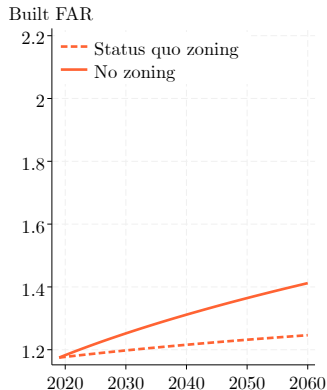


Effects of the aggregate price level

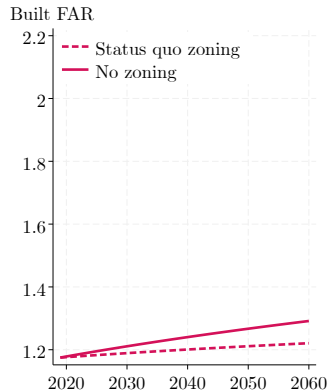
Current price level



50% lower price level (Miami)

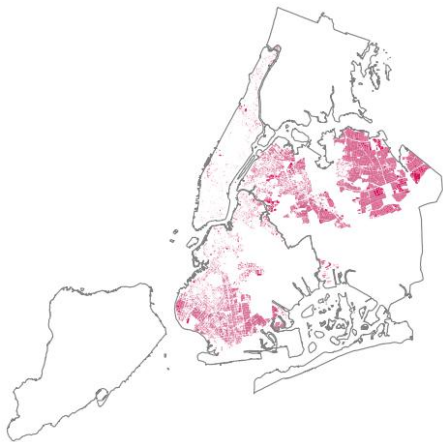


67% lower price level (Chicago)

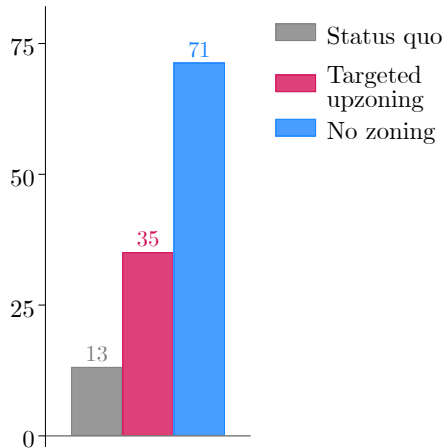


Targeted upzoning

Upzoned parcels



Floorspace growth, 2019–2060

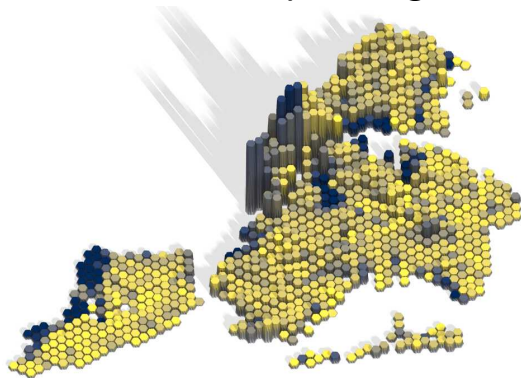
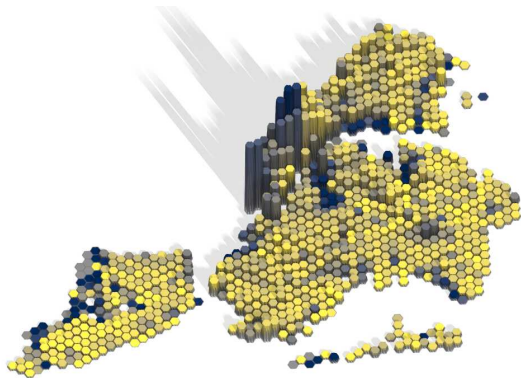


Effects on city structure

[◀ Back](#)

2019

2060, status quo zoning



Commercial share

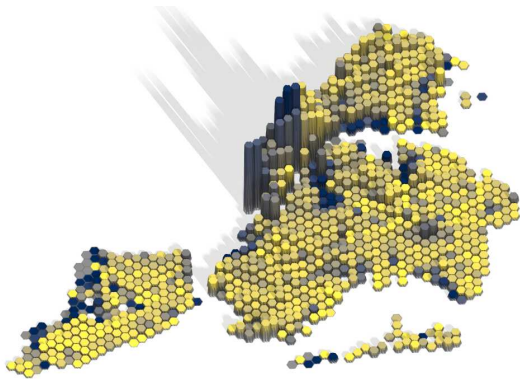


0.0 0.5 1.0

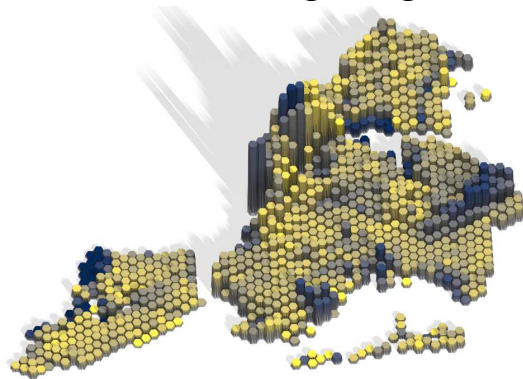
Effects on city structure

[◀ Back](#)

2019



2060, removing zoning



Commercial share



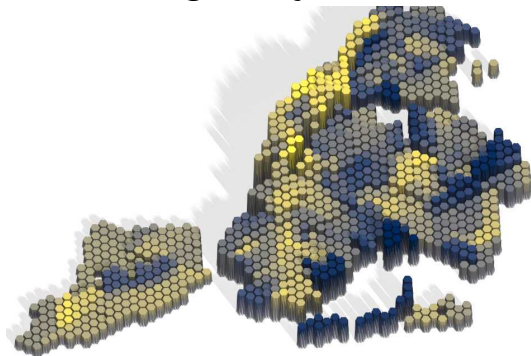
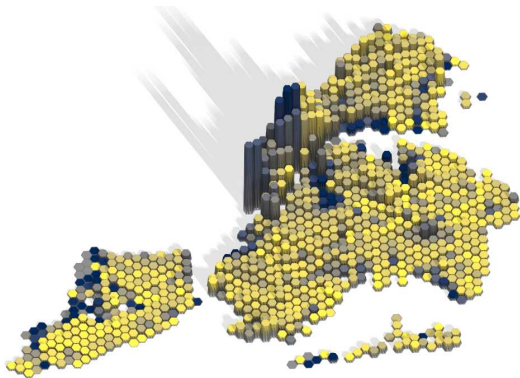
0.0 0.5 1.0

Effects on city structure

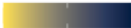
[◀ Back](#)

2019

No zoning, no adjustment cost

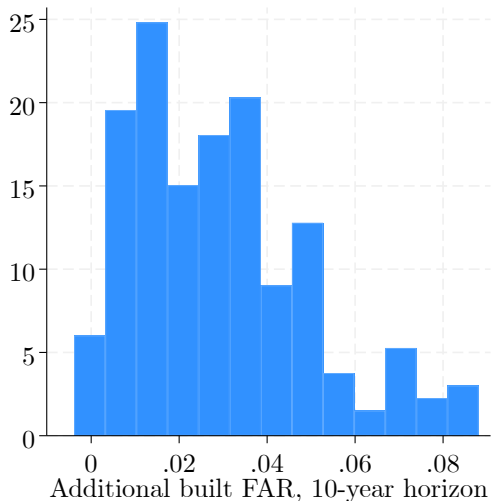


Commercial share



0.0 0.5 1.0

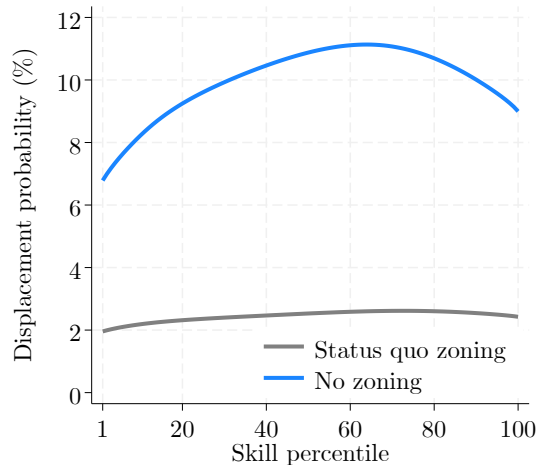
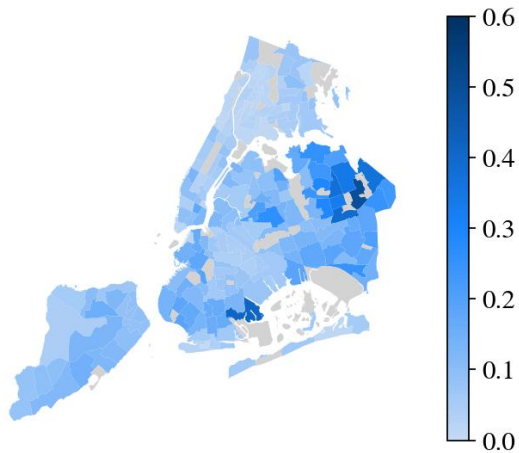
Effects of upzoning vary widely across neighborhoods



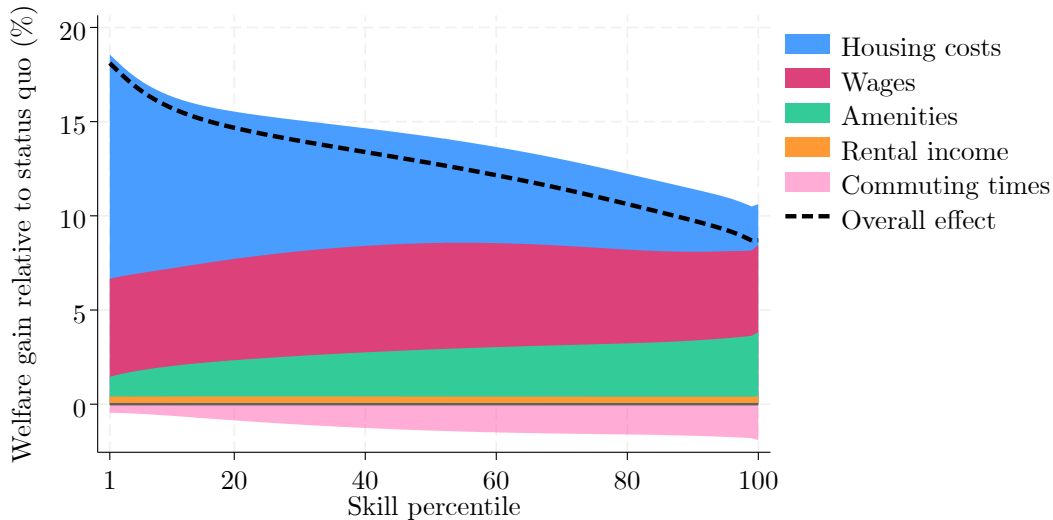
- I simulate the effect of a 1 FAR point upzoning in each neighborhood and compute FAR increases over a 10-year horizon.
 - For comparison with event study.
- Wide heterogeneity in effects.
 - Important consequences if upzoning is politically costly.
- Realized upzonings were in areas ripe for redevelopment.
 - They were “selected on gains.”

Richer households are more exposed to redevelopment

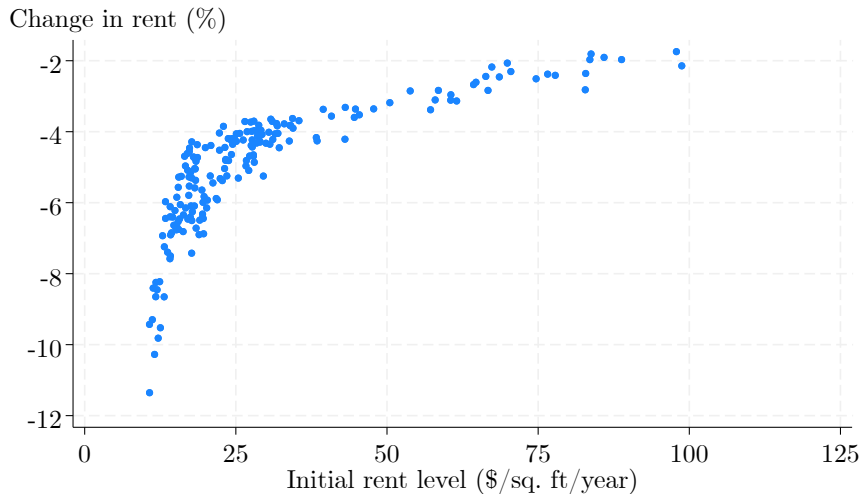
Share of land redeveloped by 2060
(no zoning regulations)



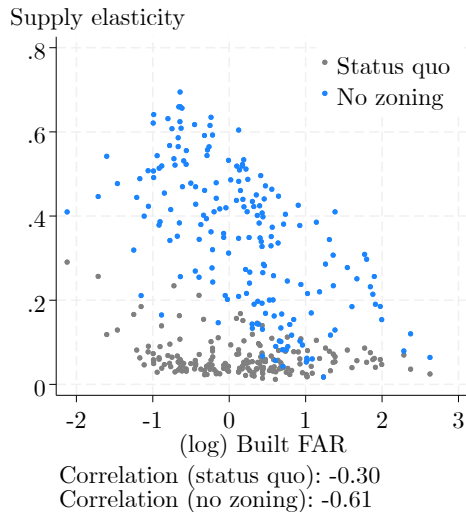
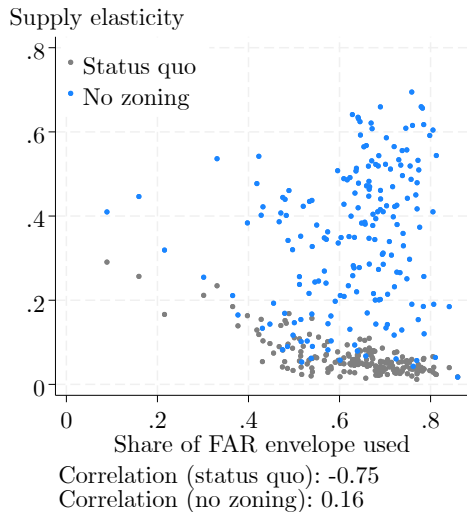
Decomposition of the welfare gains from removing zoning

[◀ Back](#)

Effects of a 10% increase in the city's floorspace on rents

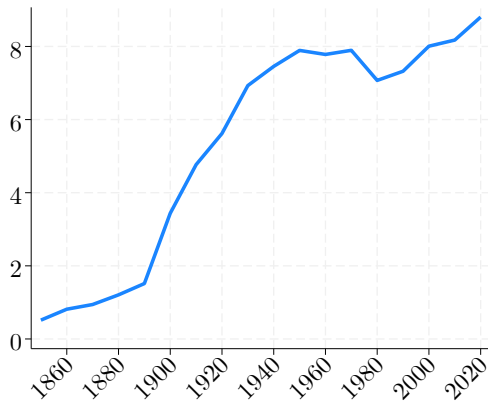


Determinants of supply elasticities

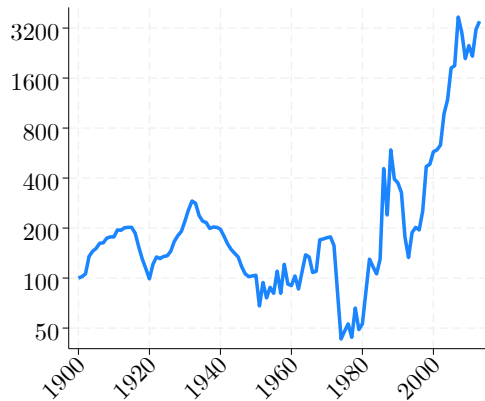


Historical population and land values

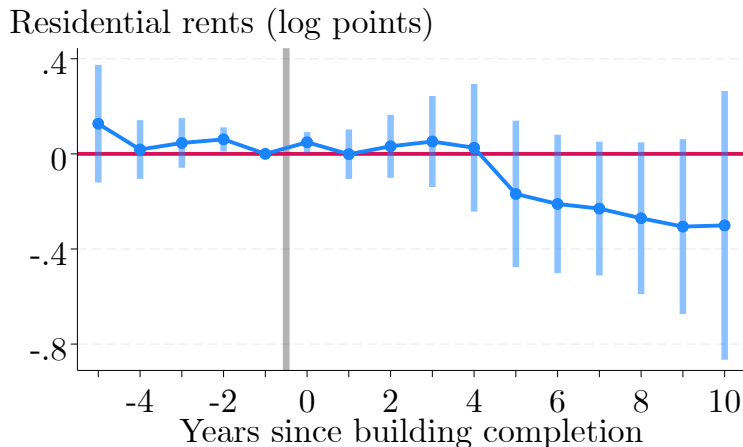
(a) NYC population (millions)



(b) Manhattan land values (1900 = 100)



Effects of industrial construction on residential rents



Manufacturing in NYC has collapsed since 1961

